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The impacts of well-connected intragranular porosity on mineral dissolution rates

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ABSTRACT

Investigations of mineral reactions in porous media are typically focused on the dynamics at water–rock interfaces, i.e., the interplay between advection, diffusion, and kinetics at grain boundaries. This interplay, however, depends on whether the solid phase is permeable. To quantify the impacts of pervious interfaces, as would be the case for grains with well-connected intragranular porosity, we conducted systematic numerical experiments of a dissolving single calcite grain with different initial internal porosities under a range of reactive transport regimes. The simulation results highlighted that the evolution of a porous grain is shaped by reactions at the interface of and within the grain. The contribution of internal reactions, depending on the rate at which reactive fluids are depleted and transported into the grain, can account for a majority of the overall dissolution. While the bulk reaction rate is enhanced due to the presence of internal reactions, the inward development of altered zones within the porous grain significantly mitigates the migration of the grain boundary. These effects caused by intragranular porosity are particularly evident in highly porous grains and under diffusion-controlled conditions. Insights gained from these numerical experiments were integrated into a hybrid conversion model, which was formulated to account for changes in grain properties (e.g., circularity, equivalent diameter, and average density) as the reaction progresses in the porous grain and is able to predict the overall reaction evolution. In presence of heterogeneity in the intragranular porosity, the overall shape evolution is retained, but grain texture varies with the extent and spatial distribution of the heterogeneity. Furthermore, the impacts of intragranular porosity on dissolution rates and the hydraulic properties were investigated in a larger domain with grain packings. The presence of intragranular porosity enhances the bulk dissolution rate by more than one order of magnitude; however, it limits the permeability increase, which is lower than that of the impervious grain packing by up to two orders of magnitude. Unlike the impermeable case, in which the permeability-porosity relationship remains consistent across reactive transport regimes, preferential surface reactions under advection-controlled conditions in packings with intragranular porosity led to a less convex porosity–permeability relationship compared to the diffusion-controlled regimes. Therefore, accounting for the effects of intragranular porosity and dissolution regimes is necessary to accurately predict the long-term reactions and evolution of porous media.

1. Introduction

Geochemically induced alterations of porous media are controlled by the interplay between flow, transport and reaction kinetics at the water–rock interfaces, and the resulting changes in the morphology and structure of the porous media will determine the evolution of their bulk properties such as porosity, permeability, and reactive surface area and thus subsequent reaction rates (Al-Khulaifi et al., 2017; Beckingham, 2017; Deng and Spycher, 2019). A mechanistic understanding of coupled pore-scale processes, especially at the fluid-mineral interfaces,

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is essential for capturing matrix heterogeneities and thus predicting bulk reaction rates and explaining phenomena such as the long-standing mineral dissolution conundrum, i.e., large discrepancies between laboratory and field reaction rates (Carroll et al., 2016; Maher and Navarre-Sitchler, 2019; Steefel et al., 2015; Zachara et al., 2016).

Recent advances in microscopic characterization techniques and pore-scale modeling have provided important insights into the dynamics of fluid–solid interfaces (Al-Khulaifi et al., 2018; Gray et al., 2021; Liu et al., 2017; Molins, 2015). Depending on the relative magnitudes of the characteristic time of advective transport, diffusive transport and kinetics, mineral reactions are characterized as surface-controlled or transport-controlled (Noiriel and Daval, 2017). The intimate coupling of different processes at the pore scale was well demonstrated by a benchmark problem involving the dissolution of a single calcite grain (Molins et al., 2021). Increasing reaction rate and elongation of the crystal shape over time were observed at high Péclet ($Pé$) and diffusive Damköhler (Da_{II}) numbers (Molins et al., 2017; Soulaine et al., 2017).

The presence of intragranular porosity further complicates the reaction dynamics. Intragranular porosity refers to micro-/nano-pores or cracks within mineral grains that are often difficult to resolve explicitly, yet is a common feature in geomaterials (Liu et al., 2020; Noiriel and Soulaine, 2021; Zachara et al., 2016). Limestones provide an example where intragranular porosity is common and important, often well-connected and contributing significantly to the overall porosity and thus the storage capacity for water or hydrocarbons (Hashim & Kaczmarek, 2019; Janjuhah et al., 2018). The micropores, typically defined as pores with diameters < 10 μm, may originate from selective dissolution of mineral phases, diagenetic alteration, intrinsic biogenic structures, or micritization and recrystallization processes (resulting in < 10 μm microcrystals also known as micrite or microspar) (Julia et al., 2023; Schultz et al., 2014). A wide spectrum of pore sizes and morphologies have been observed, and based on microcrystalline textures and their associated petrophysical properties, intragranular porosity can be grouped into three types. Type I is characterized by granular-subhedral fabrics and high microporosity (>20 %) and permeability (1–20 mD). Type II is composed of granular-euhedral and clustered textures, with microporosities between 10 % and 20 % and permeabilities of 0.1–1 mD. Type III is dominated by dense mosaic textures, and thus low microporosity (<10 %) and permeability (generally < 0.1 md) (Kaczmarek et al., 2015). Microcrystalline textures with porosities below 5 % were reported for Miocene limestones from North Sumatra, Indonesia Moshier (1989), whereas porosities exceeding 50 % were observed in the Oligocene Mount Gambier Limestone in Australia given the abundance of marine skeletal fragments and minimal burial compaction (Gharbi and Blunt, 2012). These studies demonstrated that carbonates can possess highly complex pore structures and textures, including intragranular porosity, and highlighted the need to examine the impacts of such features on fluid flow and solute transport.

From a reactive transport perspective, boundaries of mineral grains/aggregates with intragranular microstructures are permeable, and can allow mass transfer and sustain effective fluid–rock interactions, and the intragranular porosity provides a micro-environment where transport-controlled reactions can be important. Diffusion through intragranular porosity, i.e., intragranular diffusion, has been shown to have a strong impact on solute transport, especially of sorptive species (Wood et al., 1990; Ewing et al., 2010). This is because the microstructures that contribute to intragranular porosity also create surface area, which may account for 10–80 % of the BET surface area (Hudson, 1998; Hay et al., 2011) and can be accessible for reactions. As a result, reactions (e.g., denitrification and chemical weathering) have been observed to increase in presence of intragranular porosity (Lee et al., 1998; Kessler et al., 2014). Specifically, Kessler et al. (2014) showed that in permeable carbonate sediments, intragranular pores—within which diffusive transport is the dominant mechanism—create microscale anoxic niches that sustain denitrification even under generally oxic conditions, highlighting the critical role of intragranular features in regulating nitrogen cycling in carbonate-rich environments. Using Atomic Force Microscopy (AFM), Julia et al. (2023) demonstrated the formation of intragranular porosity through interface-coupled dissolution and precipitation, which is key for sustaining continued replacement of the parent calcite/aragonite grains by Cd-rich carbonate phases. Noiriel et al. (2009) examined the evolution of porosity and reactive surface area during dissolution of limestone rocks composed of micrite grains and sparite crystals. The reactive surface area—derived from Sr and Ca concentrations in the rock and outlet solution—was found to increase continuously as dissolution progressed. The observed evolution in surface area and reaction progression were not captured by X-ray microtomography (XMT) analyses, which with its limited spatial resolution (~5 μm) fails to capture the changes in surface area associated with microcrystal pores / microporosity, or by idealized geometrical models based on spherical grain-and-pore models that overlooked the enhanced reactivity caused by fluid invasion across microporous boundaries. These studies underscored the important role that intragranular porosity plays in controlling macroscopic reactive transport processes. However, systematic investigations have not been performed on the dynamics of permeable grain boundaries in the same way as it has been done for impervious mineral surfaces, or on the quantification of the impacts of intragranular porosity on evolving reaction rates.

For this purpose, it is essential to accurately capture the transition in the transport regime between the open pore space and the internal porous regions. The hybrid-scale computational approach is typically adopted, which describes fluid flow in the solid-free domain and in the continuum unresolved region using the Navier-Stokes (N-S) equation and Darcy's law, respectively, and the two regions are coupled by Beavers-Joseph boundary conditions (Molins and Knabner, 2019). While the two-domain strategy effectively captures the interfacial condition for flow, it becomes challenging when the fluid-mineral interface evolves, as the boundary conditions between subdomains need to be reconstructed and remeshed at each simulation time step. Alternatively, the micro-continuum model, enabled by the Darcy-Brinkman-Stokes (DBS) equation in a single-domain, has been used to study systems involving intragranular porosity (Scheibe et al., 2013; Kessler et al., 2014) and geosystems in general (Soulaine and Tchelepi, 2016; Soulaine et al., 2021). In this approach, the interfacial behaviors between resolved and unresolved regions are described as body source terms without the need for extra boundary conditions, and are included in a single set of partial differential equations. Consequently, this modeling framework accommodates multiscale geometrical features and moving interfaces without requiring extremely high resolution or incurring significant computational costs, which has been used for investigating the evolution of fractured porous media that also have contrasting reactive transport properties (Zhang et al., 2022b, 2024a). Thus, this approach is well-suited for simultaneously treating surface and internal reactions, and is adopted in this study to investigate impacts of intragranular porosity on reactive transport behaviors.

Moreover, for large-scale applications, it is important to develop constitutive formulations that incorporate mechanistic insights into overall reaction predictions (Ahn and Choi, 2017; Maher et al., 2006; Salmi et al., 2013; Wen and Li, 2017). For instance, in many discussions of enhanced rock weathering which involves applying rock powders on land surfaces for carbon removal, predictions of long-term mineral dissolution rate are based on the shrinking particle model (SPM) (e.g., Kelemen et al., 2020). It was initially developed in chemical engineering, along with other classical models describing fluid/gas–solid reactions, e.g., the shrinking core model (SCM) and the uniform conversion model (UCM) (Doraiswamy, 1984). These models are built on underlying assumptions and simplifications, and their applicability for geo-systems with intragranular porosity should be assessed accordingly. In both the SPM and SCM, reactions occur only on the surfaces of non-porous particles, but the SCM considers a diffusion-controlled porous ash layer resulting from the reaction of the primary phase of the particle. In contrast, the UCM assumes a homogeneous reaction

throughout the entire particle, resulting in a decreasing density while maintaining the particle diameter. These models are applicable in processes such as fuel gasification and ore leaching, but do not account for particle shape changes that can occur especially under advection-controlled conditions characterized by high *Pé* numbers (Nguyen et al., 2022). Two representative progressive models (PMs), built on the SPM and SCM to consider internal porosity, can address the formation and inward growth of an altered zone as the reaction progresses. However, the mathematical treatment for these PMs is more complex, requiring numerical integration to resolve the dissolution of porous particles (Gómez-Barea and Leckner, 2010). It is therefore crucial to have complementary numerical simulations to inform the establishment of correlations between particle properties (e.g., diameter, density, and shape) and reaction progression for developing upscaling models that are more suitable for dissolving mineral grains with intragranular porosity.

In this study, we aim to investigate the impacts of intragranular porosity, i.e., pervious interfaces, on dissolution-induced evolution of porous grains under a range of reactive transport regimes. For this purpose, we conducted a series of numerical experiments considering permeable and impermeable interfaces in a single mineral grain system, and assessed the evolution of the geometric properties (e.g., internal porosity gradient, grain shape) and overall behaviors (e.g., bulk reaction rates). These numerical simulations were enabled by an established micro-continuum modeling framework that integrates the Darcy-Brinkman-Stokes (DBS) solver in COMSOL Multiphysics (COMSOL) with the geochemical modeling capability of CrunchFlow (Section 2). This framework has been effectively used to investigate reactive transport processes in evolving altered layer bordering fractures (Zhang et al., 2022b). The simulation results were also analyzed to illustrate the impacts of different levels of intragranular porosity on grain dissolution under different reactive transport regimes (Section 3). For upscaling, a hybrid conversion model incorporating key insights from the numerical simulations was developed building on the SPM and UCM for quantifying the temporal characteristics of bulk reaction rates in porous grains (Section 4.1). The impacts of spatial variations in intragranular porosity on overall reaction rates and textural changes during grain evolution are discussed based on the numerical simulations in Section 4.2. Furthermore, the feedback between the extent of dissolution and permeability enhancement was explored at the grain-packing scale to provide upscaling insights into the impacts of intragranular porosity on porous media evolution using the micro-continuum model (Section 4.3).

2. Methods

2.1. Micro-continuum reactive transport model

Governing equations used in the micro-continuum approach are derived by cell-averaging the fundamental laws, asymptotically approaching Darcy's law in the porous matrix and the N-S equations in the solid-free space. The volume fraction of fluid (ε), i.e., local porosity, corresponding to cells containing only solid or fluid phases is 0 and 1, respectively. Grid cells with intermediate values ($0 < \varepsilon < 1$), are the fluid-mineral interface bordering the open pore space or the internal porous matrix within the grain. Micro-continuum reactive transport modeling was simulated using the COMSOL-CrunchFlow coupling framework, using MATLAB LiveLink as the application interface. A full description of the approach and numerical implementation is available in Zhang et al. (2022b, 2024a). The capability of this simulation framework to track impervious mineral surfaces at the pore scale has been confirmed based on a benchmark problem of a dissolving calcite grain. Further details of the validation are available in Text S1 (Fig. S1 and S2 in the Supplementary Material).

Specifically, flow in a medium comprising porous and non-porous regions is described by the single-field momentum conservation equation (DBS; Eq. (1) derived from the N–S momentum equation, with reactive transport solved using the advection–diffusion–reaction equation (Eq. (2).

$$\frac{1}{\varepsilon}\left(\rho_f\frac{\partial \overline{\boldsymbol{u}}_f}{\partial t}+\rho_f\overline{\boldsymbol{u}}_f\nabla\left(\frac{\overline{\boldsymbol{u}}_f}{\varepsilon}\right)\right) = -\nabla\overline{p}_f+\frac{\mu_f}{\varepsilon}\nabla^2\overline{\boldsymbol{u}}_f-\frac{\mu_f}{\boldsymbol{K}}\overline{\boldsymbol{u}}_f \tag{1}$$

$$\frac{\partial(\varepsilon\overline{C})}{\partial t} = -\nabla\left(\overline{\boldsymbol{u}}_f\varepsilon\overline{C}\right)+\nabla\left(\varepsilon D_{eff}\nabla\overline{C}\right)+R \tag{2}$$

where the averaged velocity ($\overline{\boldsymbol{u}}_f$) is defined as

$$\overline{\boldsymbol{u}}_f=\frac{1}{V}\int_{V_f}\boldsymbol{u}_f dV \tag{3}$$

physical properties of the fluid, i.e., fluid density (ρ_f) and fluid viscosity (μ_f) and the pressure field ($\overline{p}_f$) are weighted by the volume occupied by the fluid (V_f). Local permeability (K) in the porous grid cells is updated following the Kozeny-Carman permeability-porosity relationship:

$$K^{-1}=K_{const}^{-1}\frac{(1-\varepsilon)^2}{\varepsilon^3} \tag{4}$$

where the input parameter K_{const} was chosen to ensure no-slip boundary conditions at the sharp interface. For computing grids representing fluid phases, K is assigned a large value (e.g., 100) with a porosity of 1, ensuring that the DBS equation approaches the Stokes equation form. The effective diffusion coefficient (D_{eff}) in porous region is calculated using Archie's Law:

$$D_{eff}=\varepsilon^f D_0 \tag{5}$$

where D_0 is the diffusion coefficient in water and is 1×10^{-9} m^2/s, and the cementation exponent (f) was set to 1 in the micro-continuum modeling.

In our simulations, the mineral phase is calcite. Its reaction rate (R_{CaCO_3}, $mol/m^3/s$) was described by the transition state theory rate law (Molins et al., 2019):

$$R_{CaCO_3}=A\left(k_1\gamma_{H^+}C_{H^+}+k_2\gamma_{H_2CO_3^*}C_{H_2CO_3^*}+k_3\right)\left(1-\frac{Q_{CaCO_3}}{K_{sp,CaCO_3}}\right) \tag{6}$$

where A is the surface area [m^2/m^3], γ is the activity coefficient, $K_{sp,CaCO_3}$ is the solubility, and Q_{CaCO_3} is the ion activity product. k_1, k_2, k_3 are the far from equilibrium kinetic coefficients [$mol/m^2/s$]. For the cases with solid grains, A of calcite is estimated from the gradient of porosity distributions and updated as the porosity changes at the fluid–solid surface (Luo et al., 2012):

$$A=\|\nabla\varepsilon\|\psi \tag{7}$$

where $\psi=4\varepsilon(1-\varepsilon)$ is to characterize the diffuse boundary. In the porous grain cases, A within a grid cell evolves as a function of ε and the volume fraction of calcite ($V_{Calcite}$):

$$A=A_0\left[\left(\frac{\varepsilon}{\varepsilon_0}\right)\left(\frac{V_{Calcite}}{V_{Calcite,0}}\right)\right]^{2/3} \tag{8}$$

where the subscript 0 denotes the initial value. In a single-mineral system, $\varepsilon=1-V_{Calcite}$. For reactive transport simulations, ε, $V_{Calcite}$, A, K, D_{eff} were updated for each coupling time step.

To characterize the reactive transport regimes, the dimensionless Péclet (*Pé*) and diffusive Damköhler (Da_{II}) numbers are defined as $\frac{uL_c}{D}$ and $\frac{k_1\gamma_{H^+}}{DA_0}$, respectively (Soulaine et al., 2017). In this study, the characteristic length (L_c) was set to the domain width, γ_{H^+} was set to 1×10^{-3} m^3/mol following (Molins et al., 2021), and the inlet velocity (u) and the first-order rate constant (k_1) were varied to create different reactive transport conditions for numerical experiments. The Da_{II} was used instead of the advective Damköhler number because we wanted to focus on the

interplay between diffusion within grains and the reaction, and accordingly, D_{eff} (Eq. (5) was used in the determination of Da_{II}. For the kinetic term, only k_1 is included because it is the dominant term under the conditions studied here.

In the analyses of the results, we will focus on the bulk reaction rate and area-averaged reaction rate. The bulk reaction rate (mol/s) is the integral of R_{CaCO_3} taken over the volume of the grain. The area-averaged reaction rate (mol/m^2/s) is calculated by dividing R_{CaCO_3} by the reactive surface area (A, m^2/m^3).

2.2. Simulation scenario setup

Two sets of numerical experiments were conducted: the single-grain simulations were used to examine the impacts of internal porosity and reactive transport conditions on grain evolution and to develop and test a hybrid particle model for bulk reaction rate prediction; whereas the pack simulations were performed to further assess the influences on bulk reaction rates and hydraulic properties from an upscaling perspective. As illustrated in Fig. 1a, each grain is conceptualized as a simplified representation of microcrystalline aggregates (similar to micritized ooids). Further details regarding model assumptions and numerical implementation can be found in Text S2.

Simulations at the grain scale built on a well-established benchmark problem (Molins et al., 2021), which focuses on dissolution behaviors in a two-dimensional (2D) rectangular channel with a calcite grain in the middle (shown in Fig. 1b). The grain has a cylindrical shape with a radius equal to 0.1 mm, and the flow channel is 1 mm in length and 0.5 mm in width. The grain size is within the range of values reported for natural carbonate rocks, e.g., Lavoux limestone (Briolet et al., 2025) and Carrara marble (Julia et al., 2023). The domain is discretized into 250 × 125 grid cells (with a resolution of 4 μm). The computational resolution is also comparable to typical resolution of XCT imaging, meaning that microcrystals smaller than ~ 4 μm cannot be explicitly resolved. A uniform velocity boundary condition with a constant flow rate was applied at the inlet, and a fixed pressure boundary condition was used at the outlet boundary. The top and bottom walls of the channel were assumed to be non-flow, non-slip boundaries.

The flow channel was initially filled with DI water, and the influent fluid was a 0.01 M NaCl solution equilibrated with atmospheric CO_2 and has a pH of 5.0. The density (ρ_S) and molar mass (M_S) of the calcite crystal are 2,710 kg/m^3 and 0.1001 kg/mol, respectively. For calcite dissolution, k_1, k_2, k_3 are 0.89, 5.01 × 10^{-4}, and 6.6 × 10^{-7} mol/m^2/s at 25 ℃, respectively (Chou et al., 1989). Computing grids representing calcite solids were set to $\varepsilon = 0 \equiv 0.0001$, while ε was assigned values of 0.05, 0.10, 0.20, or 0.50 for porous grain scenarios considering various initial intragranular porosities. This range captures intragranular

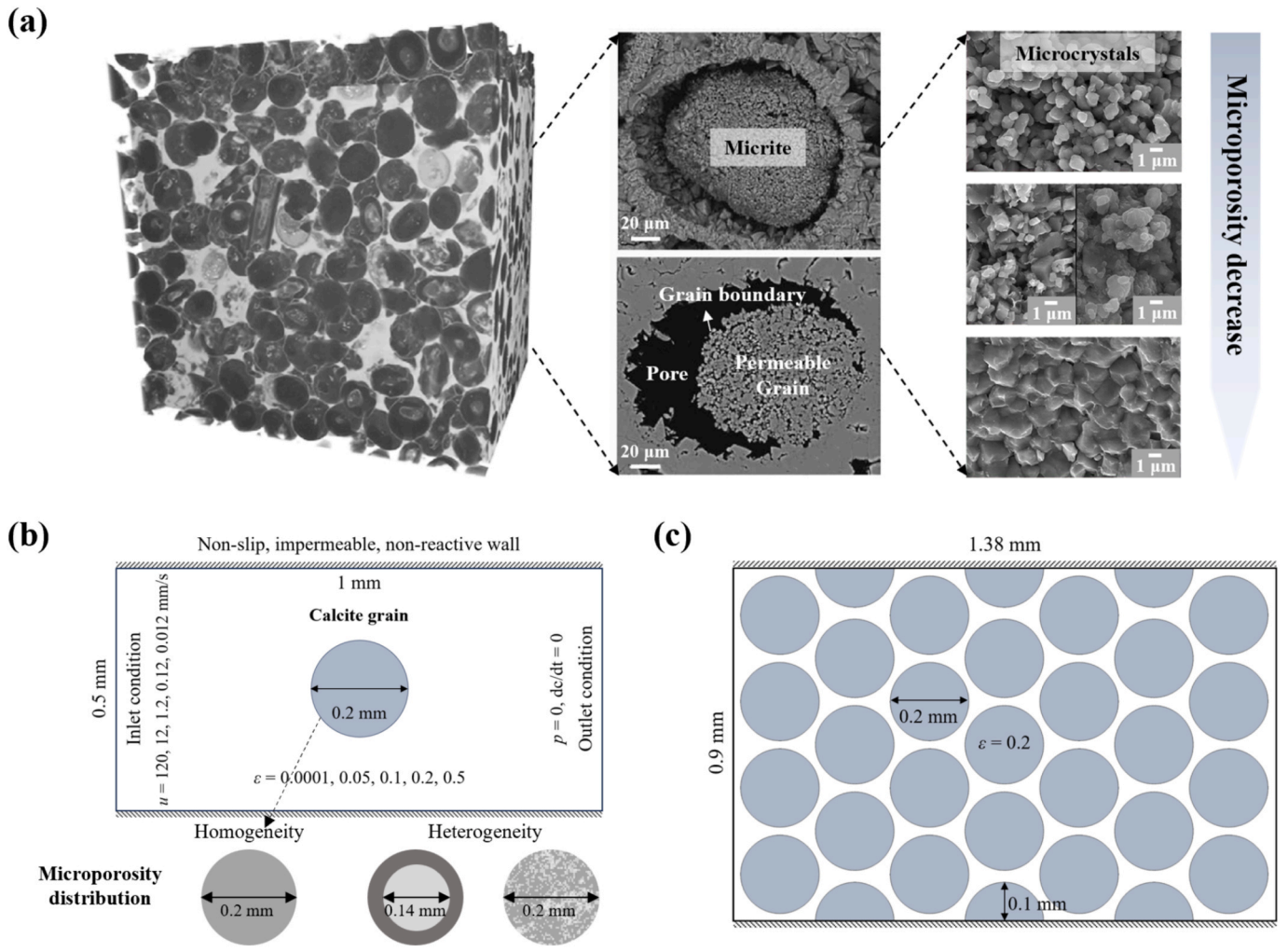


Fig. 1. Imaging of limestone pore structure and simulation setup. (a) Three-dimensional view of limestone modified after Mayo et al. (2015), with scanning electron microscopy images of micrite and microcrystals adapted from Noiriel et al. (2009) and Hashim & Kaczmarek (2019) (with permission from Elsevier). Computational domain—showing flow from left to right— along with the boundary and initial conditions for simulation scenarios, is illustrated for the (b) single-grain and (c) grain-packing sets. Spatial variations in microporosity, including both uniform and non-uniform distributions within a single grain, are illustrated in Fig. 1b. The greyscale represents various values of microporosity, and the model conceptualization is shown in Fig. S3.

porosities that are reported or used in simulations for limestones, i.e., 2–50 % (Gharbi and Blunt, 2012; Hay et al., 2011; Scheibe et al., 2013; Kessler et al., 2014). In particular, 0.1 and 0.2 correspond to threshold values from the three-tier microporosity classification (Kaczmarek et al., 2015), whereas 0.05 and 0.50 represent end-member cases (detailed in Fig. S4). Previous studies have demonstrated a log-linear relationship between porosity and permeability in microporosity-dominated limestones, and permeability values of about 0.1mD and 1 mD were reported for the porosity thresholds of 0.1 and 0.2, respectively (Hashim & Kaczmarek, 2019). It was also observed that the contribution to permeability from microporosity, when well-connected, can be comparable to that of macropores (Janjuhah et al., 2018). In this study, K_{const} was set to 1×10^{-14} m^2 in Eq. (4), which would result in permeabilities in the grains that are one order of magnitude lower than the literature values shown above, but are within the range that can be expected for realistic samples. The resulting permeability values for microporosities used in the simulations span from 0.001 to 5 mD. The specific surface area of carbonate grains can vary by several orders of magnitude depending on their composition and microstructure, ranging from approximately 0.001 to 10 m^2/g (Briolet et al., 2025; Hashim & Kaczmarek, 2019; Noiriel and Soulaine, 2021). In this study, the A_0 in Eq. 3 was set to 50,000 m^2/m^3 (equivalent to ~ 0.018 m^2/g) for permeable interface cases to ensure consistency in the initial fluid–solid interfacial area with the impervious cases.

For the baseline reactive transport regime, the $Pe = \frac{uL_c}{D} = \frac{0.012\,[m/s]\bullet 0.0005[m]}{10^{-9}[m^2/s]} = 6000$ ($\sim 10^3$), and $Da_{II} = \frac{k_1 \gamma_{H^+}}{D_{eff} A_0} = \frac{0.89[mol/m^2/s]\bullet 0.001[m^3/mol]}{0.2\bullet 10^{-9}[m^2/s]\bullet 50000[m^2/m^3]} = 89$ ($\sim 10^2$), where $u = 12$ mm/s and $k_1 = 0.89$ mol/m^2/s. In order to examine the effects of reactive transport regimes (characterized by *Pé* and Da_{II} numbers), u and k_1 were varied artificially from 120 to 0.012 mm/s and from 89 to 0.0089 mol/m^2/s, respectively. The wide range of *Pé* and Da_{II} numbers cover reactive transport scenarios of carbonate dissolution in groundwater aquifers or in geologic carbon storage systems.

Further discussion on spatial heterogeneity in microporosity within the grain is provided in Section 4.2. As shown in Fig. 1b, two types of spatial heterogeneity—layered and random distributions—are compared to further explore the impacts of fine-scale heterogeneity on grain evolution. In the layered case, the inner and outer zones have equal proportions but different microporosities, whereas the other case considers a random spatial distribution in the microporosity. The simulation conceptualization and setup for heterogeneous scenarios are detailed in Text S2 of the Supplementary Material.

An additional set of simulations was conducted to explore dissolution behaviors from an upscaling viewpoint. In a 1 mm long and 0.5 mm wide domain, multiple calcite grains are uniformly distributed in a staggered array, resulting in an intergranular porosity of 0.29 and an overall permeability of ~ 1×10^{-11} m^2. Such structural conditions are comparable to those of Ketton limestone, composed of oolitic grains (~hundreds of micrometers) with a reported porosity of 0.24 and a permeability of 3×10^{-12} m^2 (Rücker et al., 2019). To evaluate the impacts of grain intragranular pores on overall property evolution, each grain was assigned a porosity of either 0.0001 (impermeable) or 0.2 (porous), with the latter (a total porosity of 0.432) corresponding to an overall permeability of 1.19×10^{-11} m^2. A wide range of *Pé* and Da_{II} numbers (i.e., 10^2 to 10^4 and 10^1 to 10^3, respectively) were also simulated to investigate bulk reaction rates and porosity–permeability relationships under varied dissolution regimes. Other boundary and initial conditions matched those of the single-grain simulations.

3. Results

3.1. Impacts of initial intragranular porosity on porous grain dissolution

Comparisons of grain geometry and the internal structure/porosity provide critical insights into the controls of the initial intragranular porosity on overall dissolution behaviors. Fig. 2 illustrates distinct patterns of grain evolution caused by variations in the initial intragranular porosity, ranging from impervious ($\varepsilon_0 = 0.0001$) to highly permeable ($\varepsilon_0 = 0.5$) grains under the baseline reactive transport regime, i.e., *Pé*~10^3 ($u = 12$ mm/s) and Da_{II}~10^2 ($k_1 = 0.89$ mol/m^2/s). In particular, the impermeable grain exhibits more pronounced changes in grain size and shape due to the sole contribution of surface reactions, and the deceleration of flow creates a stagnant zone at the wake of the grain and a pointed shape at the downstream side over time. In contrast, internal dissolution of initially porous calcite grains results in the development of more porous and permeable zones within the grain and a slower retreat of the exterior grain boundary. When the initial intragranular porosity is low (e.g., ε_0=0.05, 0.1, 0.2), the evolution of the grain shape resembles that of the impermeable case, but the development of gradients in the internal porosity is evident. Such layered structures within individual grains are also observed in natural rocks (Batchelor et al., 2018). The stronger diffusion restriction caused by the low microporosity results in narrower but more laminations, suggesting a higher degree of local variations in reactive transport processes. In the$\varepsilon_0 = 0.5$ case, reactions occur throughout the entire grain, resulting in a more uniform mineral dissolution within the grain, which is consistent with the observations from microfluidic experiment of a highly porous gypsum grain (Dutka et al., 2020). With higher initial intragranular porosity, preferential removal of the mineral at the upstream is observed: a pointed geometry at the upgradient side develops at some point, e.g., when the remaining volume fraction is 20 % and 40 % for the$\varepsilon_0 = 0.2$ and$\varepsilon_0 = 0.5$ case, respectively, and the mass center migrates towards the downstream. Over time, the grain area with the same amount of remaining mineral in the more permeable cases can be up to five times larger than that in the less permeable cases, indicating a more limited contribution of surface reactions at the grain boundary to the overall dissolution of the grain.

Over the course of dissolution, the average bulk dissolution rates of the porous grain cases are higher than that of the impermeable grain case (4.71×10^{-10} mol/s) due to the contribution of the internal reactions. The average bulk dissolution rate is highest for$\varepsilon_0 = 0.1$ (7.84×10^{-10} mol/s), and decreases when ε_0 increases further, with a rate of 6.26×10^{-10} mol/s for$\varepsilon_0 = 0.5$. This suggests that the contribution of internal reactions does not increase monotonically with the initial intragranular porosity, because of the compound effect of faster transport at higher ε_0 within the grain and larger accessible reactive surface area but increasingly limited thermodynamic driving force towards the grain core. Meanwhile, the presence of intragranular porosity results in different trajectories of the bulk dissolution rate over time, showing an increase followed by a decrease. This concave shape in the bulk reaction rate was also observed for the dissolution of micrite carbonate rocks (Noiriel et al., 2009). At the early stage, the bulk dissolution rates increase as more surface area becomes accessible to the acidic fluids (Fig. 3b). This increase is minimum at$\varepsilon_0 = 0.5$, because when the local porosity becomes larger than 0.5, its increase would result in a lower local surface area as dictated by Eq. (3). The dominance of the surface area effect during this stage is confirmed by the observation that the bulk reaction rate and reactive surface area peak around the same time (Figs. 3a & b).

The other factor influencing dissolution dynamics is the changes in transport, as illustrated by the area-averaged reaction rate (Fig. 3c). The surface area averaged reaction rate varies by 2–3 orders of magnitude throughout time (from ~ 1×10^{-7} mol/m^2/s to ~ 1×10^{-5} mol/m^2/s) due to evolution of local transport limitations, and these values are within the range reported for slightly acidic conditions (~1×10^{-6} mol/m^2/s to ~ 1×10^{-5} mol/m^2/s; Brantley, 2008; Li et al., 2008). Overall, higher intragranular porosity enhances internal area-averaged reaction rates by enhancing reactant transport and mixing. Compared to the impervious scenario, diffusive transport is dominant within the grain, and thus the internal area-averaged reaction rate is lower than the surface reaction rate because of the longer residence time. Furthermore,

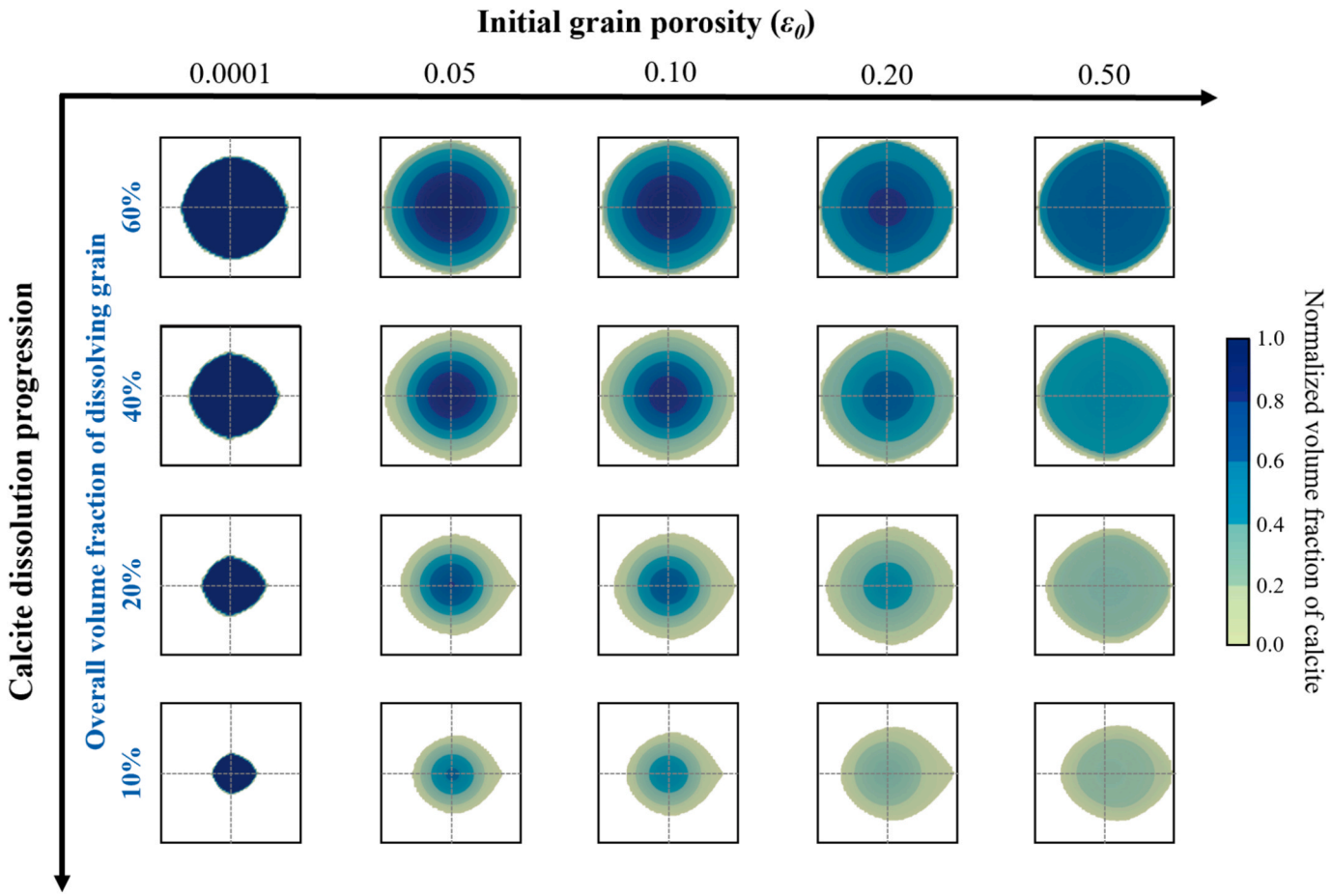


Fig. 2. Comparisons of spatial variations in calcite volume fraction across simulation scenarios with different initial intragranular porosities, ranging from impermeable ($\varepsilon_0 = 0.0001$) to highly permeable (i.e., ε_0=0.05, 0.10, 0.20, or 0.50) conditions. In these simulations, the flow proceeds from left to right at a velocity of 12 mm/s, with $k_1 = 0.89$ mol/m^2/s. The corresponding *Pé* and Da_{II} numbers are $\sim 10^3$ and $\sim 10^2$, respectively.

internal reaction would lower the chemical gradient and thus surface reaction at the grain boundary. At the later stage, transport is less limited as local porosity becomes larger, and reaction stays far from equilibrium as the remaining accessible mineral dissolution cannot deplete the reactants (see Fig. S5 in Text S3 for comparisons of spatial variations in surface area averaged reaction rate). The area-averaged reaction rate in the porous cases can exceed that of the impervious case towards the end of the grain dissolution.

Overall, the relative contribution of surface reaction is lower than that of the internal reaction under the baseline reactive transport condition (Fig. 3d). At the beginning and end of the simulations, the contribution of the surface reaction is higher, particularly for cases with lower ε_0. The ratio of surface to total reaction stabilizes at approximately 8.5 % in the intermediate stage.

In summary, the bulk reaction rates in the porous grains are larger than those of the corresponding impervious cases, with a considerable contribution from internal reactions, which is larger with higher initial intragranular porosity. The dynamics of the internal surface area and transport as the porous grain dissolves collectively control the temporal trajectories of bulk dissolution rate, area-averaged reaction rate and the grain geometry.

3.2. *Impacts of reactive transport regime on porous grain dissolution*

The interplay between advection, diffusion, and mineral reactivity can also significantly influence dissolution behaviors of porous grains. Using grains with an initial intragranular porosity of 0.2 as an example, simulation scenarios with *Pé* and Da_{II} numbers ranging from 10^0 to 10^4 and 10^1 to 10^4, respectively, were compared to evaluate the impacts of reactive transport regimes on grain shape and dissolution rate.

We observed three distinct types of exterior shapes. Under the advection-controlled condition with high reaction kinetics, characterized by a high *Pé* number and Da_{II} number, the grain boundary is featured by an elongated shape that is more pointed at the downgradient side with minimum changes in the internal porosity (e.g., Type I shown in Fig. 4). The more effective transport in the open pore space and higher reactivity (e.g., Type I) result in much higher overall bulk dissolution (Fig. 5a). The contribution of surface dissolution is dominant (Fig. 5d) and thus the grain geometry shows similar behaviors as those of impermeable grains at the higher flow velocity (depicted in Fig. S1 in the Supplementary Material). With moderate *Pé* and Da_{II}, the grain is slightly elongated and more pointed at the upgradient side (Type II, highlighted in Fig. 4), demonstrating considerable internal reactions near the upstream region as facilitated by the pervious interfaces. The contributions of internal reactions are also evidenced by the development of internal porosity gradient. When diffusion is dominant (i.e., at low *Pé*), reaction is more uniform on all exterior surfaces, allowing the grain to retain a perfect cylindrical shape (Type III). This uniform surface reaction is similar to the impermeable case at a slowest velocity (Fig. S1), highlighting the influence of flow regimes on surface evolution (Molins et al., 2017). However, contributions of surface reaction are limited (Fig. 5d). Noticeable shrinkage only occurs until the altered zones become extremely porous. Additional heterogeneity within the grain is also observed under the diffusion-controlled transport conditions (e.g., Type IV with *Pé*~1), featured by the movement of the mass center to the downstream. Although the shape remains cylindrical,

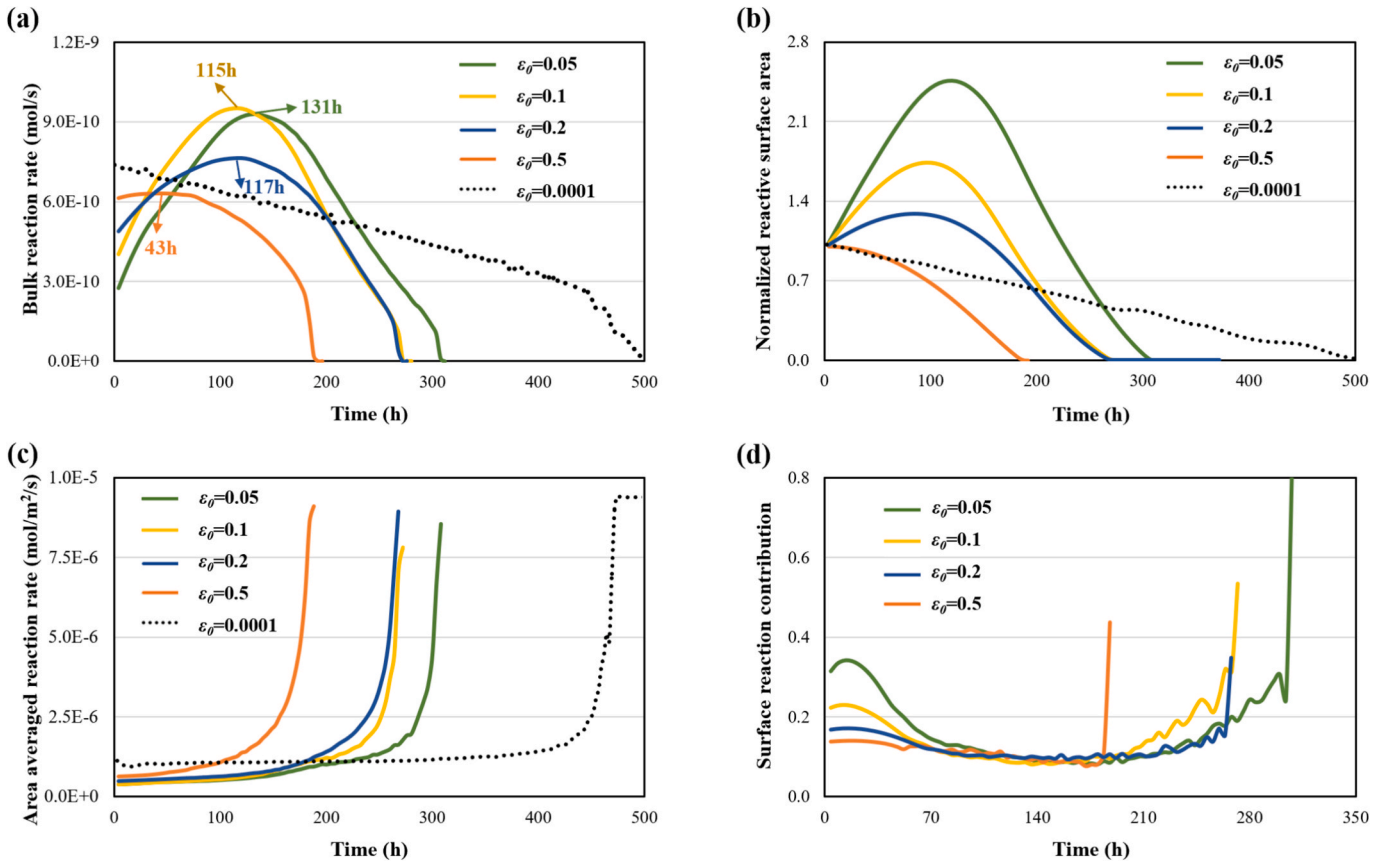


Fig. 3. Variations in (a) bulk reaction rate, (b) normalized reactive surface area (i.e., bulk surface area normalized against the initial value), (c) area-averaged reaction rate, and (d) relative contribution of surface reactions over time across simulation scenarios with different initial intragranular porosities. Colored arrows in Fig. 3a indicate the time points of maximum reaction rates. Surface reaction contribution is determined by the ratio of surface reaction rate (mol/s) to bulk reaction rate (mol/s). All simulation scenarios were conducted under the baseline reactive transport regime ($Pé$~10^3, Da_{II}~10^2), corresponding to the Type II regime described in the following discussion.

indicating uniform surface reaction, the internal reaction is more effective at the upstream as transport of the reactants downstream is difficult. In this regard, simulation-based investigations offer valuable insights into the morphology and internal layering of mineral grains, which can be used to inform the reconstruction of the hydrodynamic environment during their formation.

As $Pé$ and Da_{II} numbers decrease (from Type I to Type IV), the average bulk dissolution rate decreases from 1.60×10^{-9} mol/s to 3.35×10^{-10} mol/s (Fig. 5a), the area-averaged dissolution rate (Fig. 5c) and the contribution of surface reaction (Fig. 5d) are higher, except for the case corresponding to Type IV. At higher $Pé$ and Da_{II} numbers (e.g., Type I), grain evolution is predominantly surface-reaction-driven, with the ratio of surface reaction rate to bulk reaction rate exceeding 0.6. The initial increase in the bulk reaction rate is attributed to the more effective transport and higher area-averaged reaction rate, whereas the later decrease is a result of the decrease in the accessible reactive area. At lower $Pé$ and Da_{II} numbers, as detailed in the previous section, the temporal trends are controlled primarily by the surface area, and the contribution of internal reactions can reach 92 % of the overall dissolution. In this regard, incorporating the influence of internal reactions driven by intragranular porosities into reactive transport models is critical for accurately predicting the overall dissolution and its dynamics, particularly under diffusion-dominant conditions.

4. Discussion

4.1. A rate model for overall reaction prediction

In order to evaluate overall behaviors of evolving single grains, we also adopted the general concept of the shrinking particle model with modifications that consider the porous nature of the grain. For the dissolution of impervious crystals free of any insoluble layer around the particle, and the surface reaction process can be described by the SPM (as illustrated in Fig. 6a). The changes in equivalent diameter (d) and density ($\bar{\rho}$) of evolving particles are given by the following expressions (Nguyen et al., 2021):

$$\frac{d}{d_0} = (1 - x_S)^{\alpha} \tag{9}$$

$$\frac{\bar{\rho}}{\rho_0} = (1 - x_S)^{\beta} \tag{10}$$

where d_0 is the initial diameter of particles, ρ_0 is the initial density, and x_S is the mass fraction of the reacted solid. α and β are exponents that depend on particle properties and reaction regimes. For a cylindrical grain, $\alpha = \frac{1}{2}$ and $\beta = 1 - 2\alpha$, i.e., $\beta = 0$, and the density of the dissolving grain remains constant.

For porous grains, reactions occur at the grain surface and within the

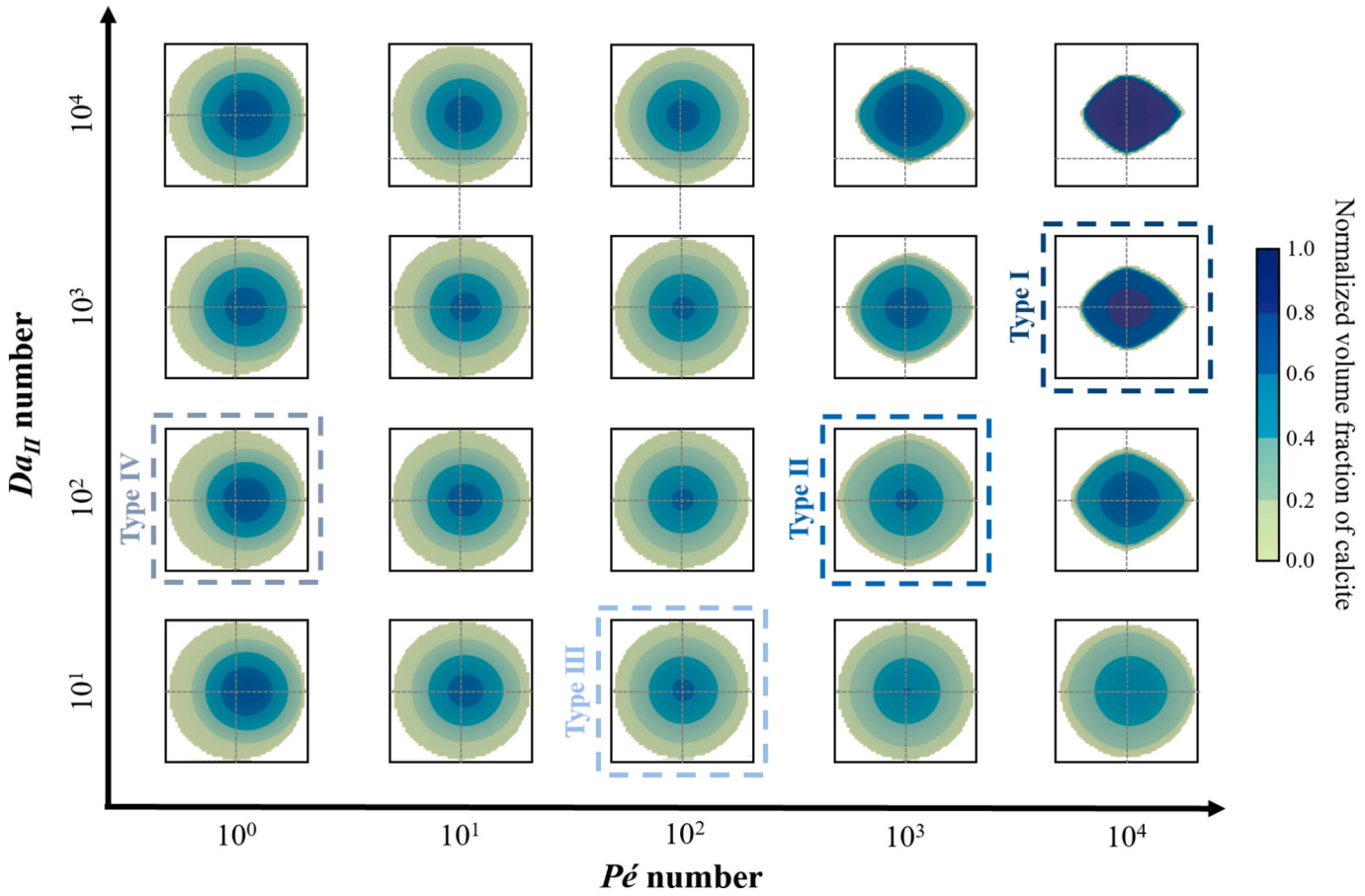


Fig. 4. Dissolution pattern diagram of porous grains ($\varepsilon_0 = 0.2$) according to varying *Pé* and Da_{II} numbers. Time snapshots are selected for each scenario when the total volume fraction of calcite grains reaches 30 % with the flow from left to right. Regarding grain geometry and porosity distribution, four dissolution types are highlighted for further discussion.

grain, resulting in an increasingly porous layer developing towards the core, and thus a reduction in both diameter and density (as shown in Fig. 6b). Such behaviors can be described using the PM, but the variations in parameters α and β as the reaction progresses are rarely addressed in current PMs (Halama and Spliethoff, 2015; Ma, 2006). Here, we propose a hybrid particle model that combines the concepts of SPM and UCM. Assuming uniform internal reactions, and that reactant concentration and matrix density are homogeneously distributed within the dissolving grain (Fig. 6b). It should be noted that unlike the micro-continuum model which uses the transition state theory-based reaction rate, development of the hybrid conversion model uses the reaction rate law with a first-order dependence on the concentration of H^+ for simplicity. Inter-model comparison under identical geochemical conditions confirmed that both the transition state theory rate law and a first-order reaction rate law can reproduce the experimental observations (Molins et al., 2021).

Specifically, at the grain boundary, mass conservation dictates that the transport rate of reactant through the boundary equals the consumption of the reactant at the boundary as in the SPM and by the internal reactions as in the UCM (Nguyen et al., 2022; Raschman et al., 2019). Here, the mass transfer rate of H^+ ($\frac{dN_{H^+}}{dt}$, [mol/s]) between the bulk solution and the grain surface equals the consumption of H^+ within the porous grain by calcite dissolution, giving:

$$\frac{dN_{H^+}}{dt} = k_L P\left(C_{bulk} - C_{gav}\right) \tag{11a}$$

$$\frac{dN_{H^+}}{dt} = Ak_1\gamma_{H^+}C_{gav}S \tag{11b}$$

where P, S, C_{bulk} and C_{gav} are the shrinking grain perimeter, grain area, H^+ concentrations in the bulk solution, and the grain-averaged concentration within the calcite grain, respectively. Herein, the bulk concentration is kept practically unchanged during the process (i.e., $C_{bulk} \cong C_0$). The mass transfer coefficient (k_L) can be calculated from the Sherwood number (Sh), which is related to the grain Reynolds number ($Re = \frac{u\rho_f L_c}{\mu}$) and the Schmidt number ($Sc = \frac{\mu}{\rho_f D}$) (Nguyen et al., 2022):

$$k_L = \frac{ShD_{eff}}{r} \tag{12}$$

$$Sh = 2 + 0.6Re^{\frac{1}{2}}Sc^{\frac{1}{3}} \tag{13}$$

where ρ_f and μ are the density and dynamic viscosity of the fluid, respectively. By solving Eq. (11a) and (11b) for the unknown C_{gav} and substituting it back, mass transfer rate can be determined by:

$$\frac{dN_{H^+}}{dt} = \frac{k_L P C_0 A k_1 \gamma_{H^+} S}{(k_L P + A k_1 \gamma_{H^+} S)} \tag{14}$$

Then, the overall reaction rate of a porous grain is expressed as:

$$\frac{dx_S}{dt} = \frac{d(m/m_0)}{dt} = \frac{dN_{H^+}}{dt}\frac{M_S}{nm_0} = \frac{M_S C_0 k_1 \gamma_{H^+}}{2m_0}\frac{k_L APS}{(k_L P + A k_1 \gamma_{H^+} S)} \tag{15}$$

where m and m_0 are the reacted and inital mass of the grain, respectively, and M_S is the molar mass. The stoichiometric coefficient n is 2 for calcite dissolution. In this work, a shape exponent (φ) was added to the perimeter function, i.e., $P = \pi d(1 - x_S)^{\varphi}$, to describe the variations in

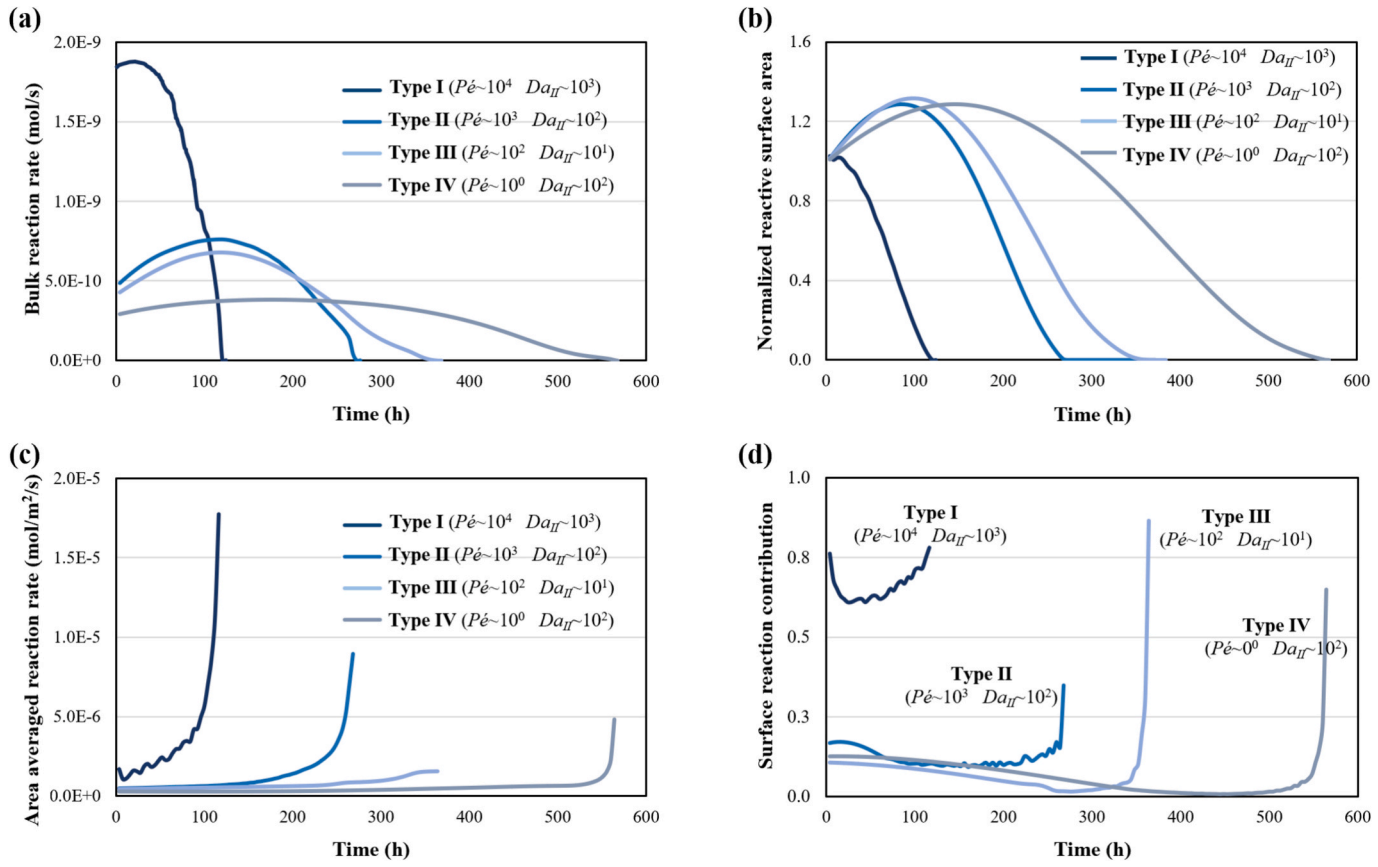


Fig. 5. Variations in (a) bulk reaction rate, (b) normalized reactive surface area, (c) surface area averaged reaction rate, and (d) relative contribution of surface reactions over time over time for different dissolution types selected based on Fig. 4.

grain circularity during reaction, while $S = \frac{1}{4}\pi d^2$ always holds by definition. By incorporating parameters α, β and φ into the proposed hybrid model, the impacts of initial intra-granular porosity and dissolution regimes on the dynamics of grain size, density and shape, and thus overall reaction rate can be assessed.

The numerical simulations provide detailed information for evaluating φ, α and β. Fig. 7 displays the temporal profiles of features such as equivalent diameter and circularity of the grain for cases with different intragranular porosity and reactive transport regimes. The slopes of the ln-ln plots correspond to the exponents α and φ, with higher values indicating a larger change during reaction. For the impervious case ($\varepsilon_0 = 0.0001$), where reactions only occur at the fluid–solid interfaces, high values of α and φ demonstrate evident shrinkage and elongation of the grain as dissolution progresses. In the permeable cases, simultaneous contributions of surface and internal dissolution result in three distinct stages as identified by the varying slopes shown in Fig. 7a. In Stage I, internal dissolution and inward alteration of the grain dominates, leading to limited variations in grain size and shape (characterized by constant diameter and circularity) across all permeable scenarios. In the second stage, the internal and surface reaction are both in play, resulting in continuous and evident change in grain circularity and diameter (steeper slopes in Figs. 7a and 7c). During this period, the grain elongates and shrinks due to the contribution of surface reactions, but to a less extent compared to the impervious cases (confirmed by lower values of α and φ). As the reaction progresses towards the end (Stage III), α and φ increase significantly due to the rapid elimination of the extremely porous zones as the grain shrinks. It is important to note that the correlation between α and β (i.e., $\beta = 1 - 2\alpha$) remains valid in porous grains, despite the presence of the complex dynamics.

Generally, α changes significantly during dissolution, and an additional function is defined as follows:

$$\alpha = 0.5e^{\alpha_1(1-x_S)} \tag{16}$$

where α ranges from 0 to 0.5, and α_1 is a constant that relates to the rate of grain dissolution. As an indicator of shape circularity, φ exhibits a near-linear correlation with the natural logarithm of reaction (Kopanja et al., 2016) and displays varying slopes depending on dissolution patterns. For a cylindrical grain, as discussed in this work, the exponent β can be replaced with $1 - 2\alpha$. Inputting the fitted shape parameters (e.g., Eq. (16) in the hybrid model (Eq. 9–15), the overall mass fractions were reproduced using the proposed hybrid conversion model and compared to the simulation results (shown in Fig. 8). The fitted values of α_1 and φ are summarized in Table 1. All R-squared values exceed 0.999, confirming the hybrid model's potential in describing the dissolution of porous grain across various dissolution patterns.

Consistent with the grain size and shape across scenarios, the parameters α_1 and φ are more sensitive to the reactive transport regimes than to the initial internal porosities. A smaller α_1 indicates delayed shrinkage and a higher φ corresponds to delayed shape change, and they are typically correlated. Particularly, transitioning from a tight matrix ($\varepsilon_0 = 0.05$) to a highly porous one($\varepsilon_0 = 0.5$), α_1 increases by 40 %, while φ decreases by 43 %, highlighting the enhanced contributions of internal reactions. Similar changes in fitted parameters are observed for the reactive transport regime with $Pé \bullet Da_{II}$ that's four orders of magnitude lower, but to a larger extent. Multiple regression analyses reveal a strong negative correlation between φ and $\log_{10}(Pé \bullet Da_{II})$ with a P-value of 0.03, while α_1 is less influenced by reactive transport conditions. The extent of intragranular porosity has a more limited effect on α_1 and φ (see Fig. S6 in Text S4 for further correlation analyses of the parameters). Even in tight rocks (e.g., porosity lower than 5 %), a shift from surface-reaction-dominated evolution to internal reactions is expected

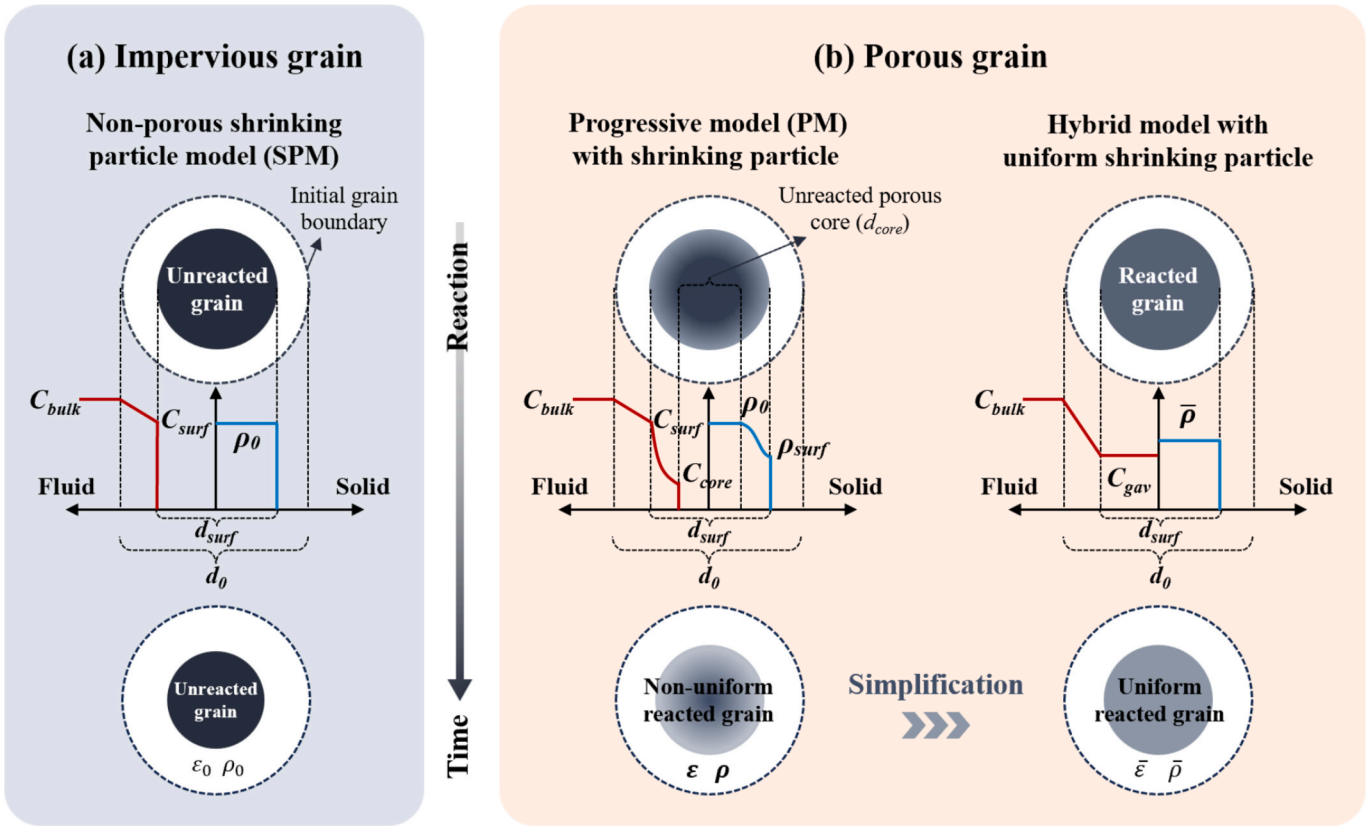


Fig. 6. Schematic illustration of altered grain structures according to different model assumptions for both (a) impervious and (b) porous cases. Herein, the subscripts *bulk*, *surf*, *core*, and *gav* denote bulk solution, grain surface, unreacted core surface and grain averaged value, respectively.

as a more porous and thicker altered zone develops over time, delaying the evolution of mineral surfaces.

While SPM and UCM have been widely used in describing the individual contribution of surface and internal reactions to grain evolution, respectively, they may lead to significant inaccuracies in predicting overall reaction rates for shrinking porous grains under reactive regimes when both contributions are significant. Therefore, the proposed hybrid model, which accounts for the coexistence of surface and internal reactions, broadens the applications of this type of model in predicting overall dissolution rates in a multitude of systems, including rare earth element recovery, mineral carbonation, hydrometallurgy, and hydrate formation (Gerardo et al., 2023; Zhang et al., 2024b; Popovič et al., 2020; Zhang et al., 2022a).

4.2. Spatial heterogeneity in microporosity

As we conceptualized the grain as an assemblage of microcrystals, it needs to be recognized that microcrystal-scale heterogeneity in size and spatial arrangement can introduce additional complexity as observed in natural systems. We further explore the impacts of microcrystal-scale heterogeneity on overall grain evolution by varying internal porosity, surface area, and permeability according to different microcrystal sizes, packing modes, and spatial assemblage patterns (layered vs. mixed heterogeneity).

For microcrystals of uniform size, packing geometry alone can cause porosity to vary significantly—from a theoretical maximum of 47.6 % in cubic packing (CP) to 26.0 % in hexagonal close packing (HCP). As microcrystal size (d) varies, the associated specific surface area (A) and permeability (K) can be updated by $A = \frac{6(1-\varepsilon)}{d}$ and $K \propto \frac{d^2\varepsilon^3}{(1-\varepsilon)^2}$, respectively. Furthermore, mechanical compaction or cementation can bring microcrystals into closer contact, progressively reducing pore space and permeability—a condition referred to as sutured contact packing (SCP).

Five microcrystal-scale heterogeneities were considered to reflect alternating placement of coarser vs. finer microcrystals (Layered_size1/2) and different packings (HCP vs. SCP, Layered_pack1/2). Meanwhile, the Mixed_1 case represents a randomized microcrystal-scale heterogeneity, offering a direct comparison to the uniform scenario with the same total porosity. Additional details on the simulation setup for each heterogeneous scenario can be found in Table S1.

Fig. 9 illustrates how initial heterogeneities influence overall reaction rates and spatial dissolution patterns within the grain. A low-permeability outer layer (as in Layered_size2 and Layered_pack2) acts as a diffusion barrier, limiting accessibility into the grain and reducing dissolution at early stages. This configuration results in the development of a porous ring/crust surrounding a shrinking core. This structure captured by the simulations is consistent with features previously documented in carbonate rock samples, as also evidenced in Fig. 1a (Gharbi and Blunt, 2012; Noiriel et al., 2009). Conversely, when the outer region is more permeable—as in the cases of Layered_size1 and Layered_pack1—it is preferentially dissolved, leaving behind a denser core with slower reactions. The subsequent dissolution pattern resembles that in Fig. 2 for grains with uniformly low initial intragranular porosity, but shows limited variations in the gradient of the internal porosity. In the mixed heterogeneity scenario, randomly distributed microcrystal sizes, spatial arrangements, and thus local properties, tend to reduce overall diffusion from the grain surface toward the interior and produced a more compact reacted core compared to the homogeneous case. But the overall reaction rate is faster. Fine-scale variations also result in a rougher boundary despite retaining a similar overall grain shape compared to the reference case.

In summary, while the overall shape conforms to the four types illustrated in Fig. 4, microcrystal-scale heterogeneities as can be induced by variations in microcrystal sizes, packing modes, and spatial arrangements (either layered or mixed) can give rise to different fine-scale

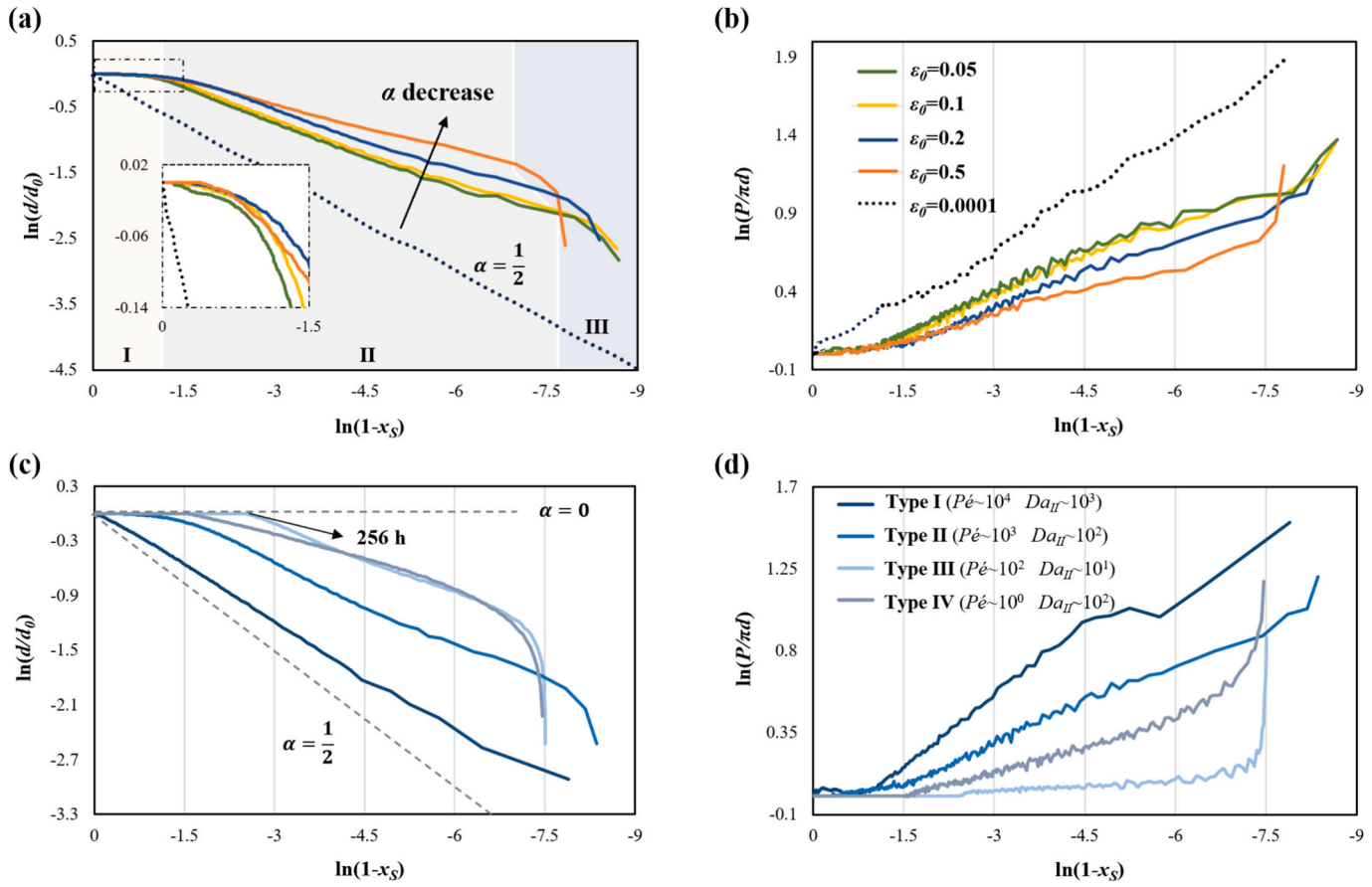


Fig. 7. Dependence of (a, c) grain equivalent diameter and (b, d) grain circularity on the reaction progression across simulation scenarios with different (a, b) initial porosities and (c, d) reactive transport regimes. In particular, the Type II transport regime was used for the simulations shown in Figs. 7a and 7b, and an initial porosity of$\varepsilon_0 = 0.2$ was set for the simulations in Figs. 7c and 7d. The exponents α and φ correspond to the slopes of ln-ln plots shown in Figs. 7a and 7c, Figs. 7b and 7d, respectively. Three stages of reaction process highlighted in Fig. 7a, (i.e., I, II, and III) can be identified based to the dynamics of α.

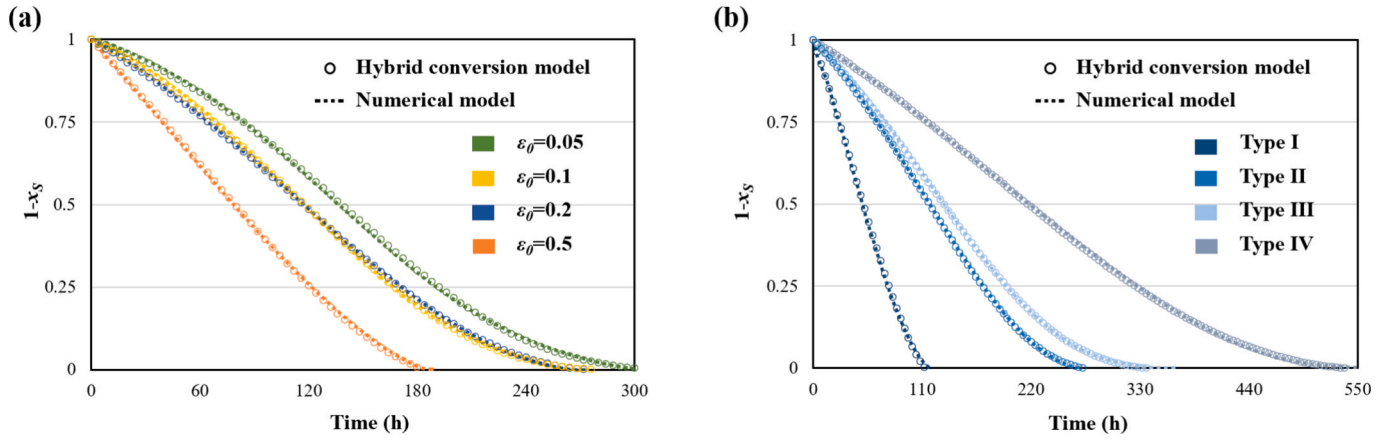


Fig. 8. Comparisons of remaining mass fraction curves from numerical model (micro-continuum approach) and hybrid conversion model across scenarios with varying (a) initial grain porosities and (b) reactive transport regimes.

textural features (e.g., the thickness and porosity of the internal layers) during dissolution. New features, such as porous rings surrounding an inner core—caused by diffusion barriers in the outer layer—or irregular boundaries as shaped by localized heterogeneity, may develop.

These features are also observed in natural carbonate grains. For example, carbonate ooids are typically characterized by concentric laminae surrounding a central nucleus. Although multiple hypotheses have been proposed for their genesis—including physical agitation, abiotic and microbial-induced mineral precipitation—a universally accepted model remains elusive (Batchelor et al., 2018). Our simulation results suggest that a dissolution-driven mechanism – a preferentially shrinking core encased by a more stable outer ring – may also be involved. It may be coupled with other mechanisms such as diagenetic recrystallization, and lead to the progressive development of concentric laminae. Although these discussions focus on carbonates, the perspective gained should equally apply to ooids mineralized by chamosite,

Table 1
Parameters α_1 and φ for the hybrid conversion model fitted based on simulation scenario results.

Parameter	ε_0	*Pé*	Da_{II}	α_1	φ	R^2
Scenario						
Scenario						
Initial grain porosity	0.05	10^3	10^2	−0.90	−0.014	0.9996
	0.1	10^3	10^2	−1.18	−0.011	0.9997
	0.2	10^3	10^2	−1.24	−0.010	0.9997
	0.5	10^3	10^2	−1.26	−0.008	0.9999
Reactive transport regime	0.2	10^4	10^3	−0.39	−0.020	0.9998
	0.2	10^3	10^2	−1.24	−0.010	0.9997
	0.2	10^2	10^1	−1.65	−0.001	0.9998
	0.2	10^0	10^2	−1.27	−0.007	0.9998

stevensite, phosphate and possibly some other mineral phases.

4.3. Physicochemical property evolution

Investigations of the single grain evolution highlighted the impacts of permeable fluid-mineral interfaces on the dynamics of grain geometry and reaction progress under different initial intragranular porosity and reactive transport conditions. Grain geometries and reaction extent will impact the overall evolution of the properties of the matrix composed of such dissolving grains. To provide an upscaling perspective, numerical experiments were conducted at a larger scale to evaluate spatial variations in porous grain dissolution within uniformly packed grain assemblies.

Fig. 10 illustrates the spatial patterns of reacted packings of porous grains across simulation scenarios with *Pé* and Da_{II} ranging from 10^2 to 10^4 and 10^1 to 10^3. Consistent with the single-grain simulations, enhanced advective transport and mineral reactivity lead to more evident grain shrinkage and limited internal alterations. For relatively high *Pé* and Da_{II} numbers (Fig. 10a), the flow pathways between porous grains are enlarged significantly across the entire domain, exhibiting a spatial pattern comparable to the impervious case (Fig. 10d). This is in contrast with the cases dominated by internal reactions (Fig. 10c). A comparison of spatial patterns between Figs. 10b, 10c and 10d highlight the limited widening of flow paths induced by permeable fluid-mineral interfaces under these reactive transport conditions. The extent of dissolution generally decreases along with the flow direction, as evidenced by the remaining volume fractions for each grain column. The minimum value serves as an indicator of the reaction front, which penetrates deeper in the advection-controlled transport regime resulting from further surface reactions (Fig. 10a).

Variations in dissolution behaviors caused by initial porosities and reactive transport regimes can lead to distinct evolution trends for hydraulic properties. Fig. 11 illustrates the overall reaction progress and corresponding permeability changes in porous media across the simulation scenarios. In scenarios dominated by surface reactions, the overall dissolution rates consistently decrease over time. In contrast, scenarios characterized by relatively low *Pé* and Da_{II} numbers exhibit increasing dissolution rates during the early and intermediate stages (Fig. 11a). This suggests that internal reactions contribute more significantly to overall dissolution due to enhanced transport within dissolving grains. However, higher reaction rates do not necessarily correspond to pronounced permeability increases. Permeability enhancement is more substantial in scenarios with impervious grain surfaces, even though their average reaction rates are lower than those of scenarios with porous grains (Fig. 11b). When grain shrinkage is limited, as shown in Fig. 10, the widening of flow paths within the grain packings is minimal, thereby constraining permeability enhancement in the domain. A further decrease in the *Pé* number does not introduce a distinct pattern

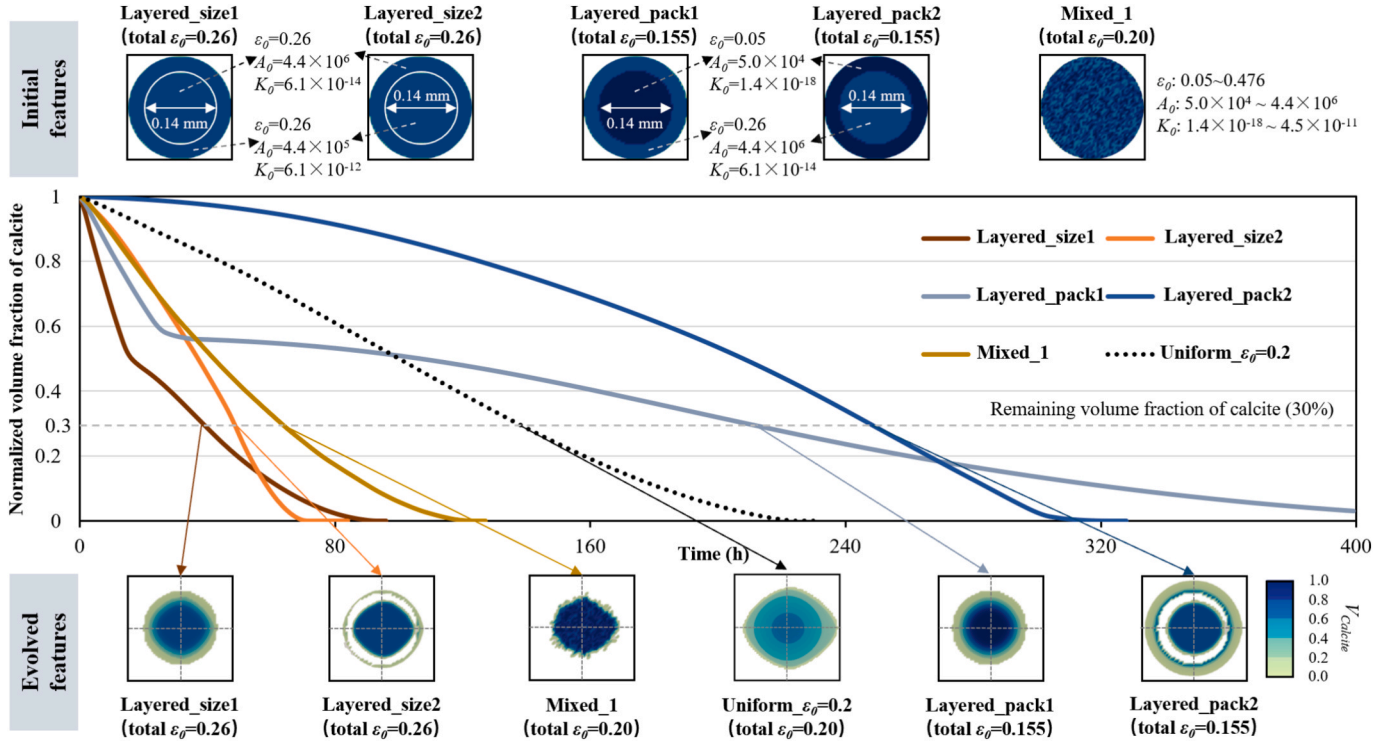


Fig. 9. Comparisons of calcite volume fraction evolution and spatial variations across simulation scenarios considering microcrystal-scale heterogeneities (Layered_size1/2: inner layer – fine/coarse microcrystal; Layered_pack1/2: inner layer – SCP/HCP). In these simulations, the flow velocity is 12 mm/s, the k_1 is 0.89 mol/m^2/s, and the *Pé* and Da_{II} are ~ 10^3 and ~ 10^3, respectively. The surface area is calculated from $A = \frac{6(1-\varepsilon)}{d}$. The uniform case ($\varepsilon_0 = 0.2$) serves as a reference under the same *Pé* and Da_{II} conditions. Time snapshots in the bottom panel are taken when the normalized calcite volume fraction reaches 30 % in each scenario.

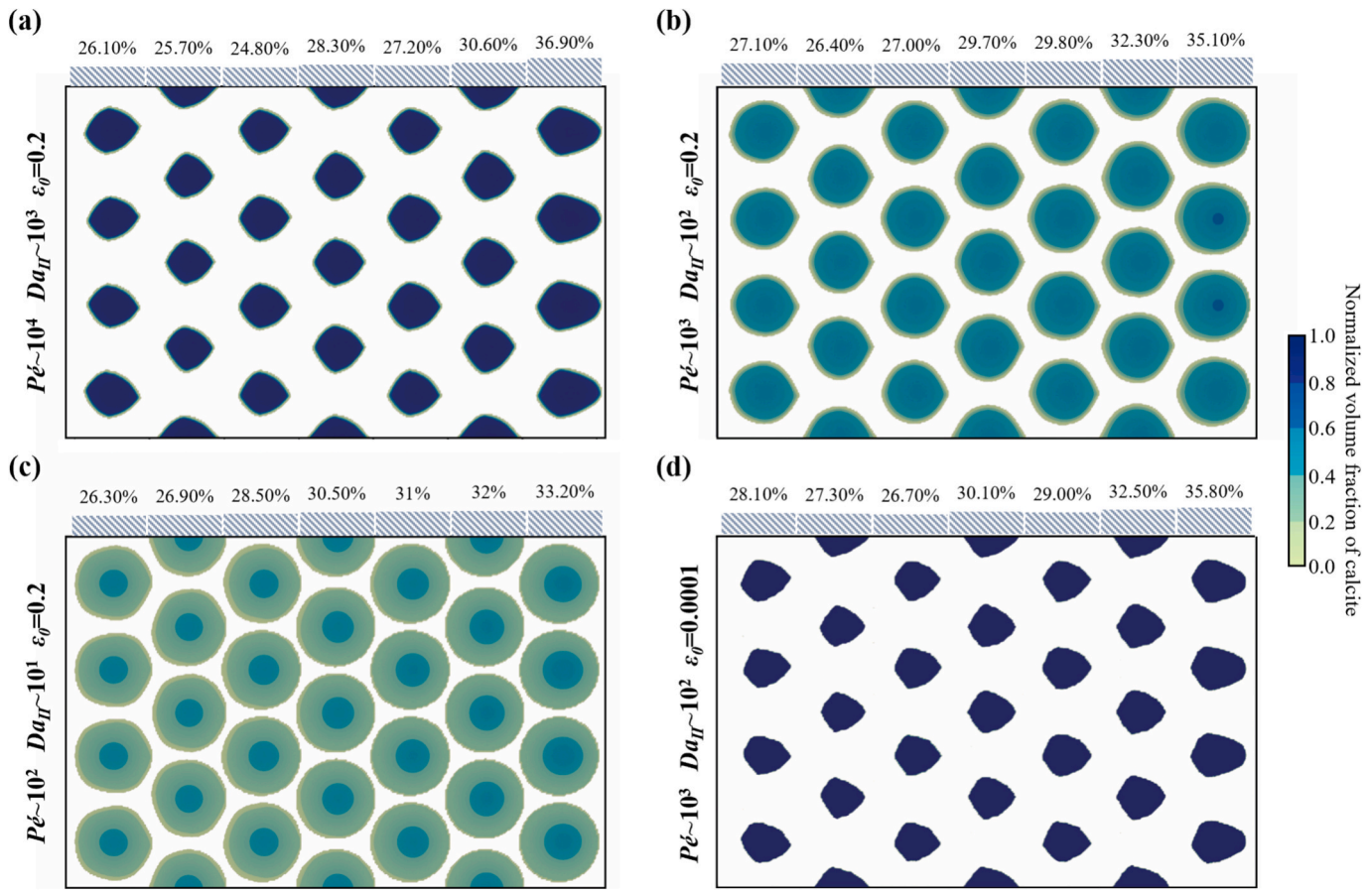


Fig. 10. Comparisons of spatial variations in calcite volume fraction across simulation scenarios with different reactive transport regimes and initial intragranular porosities. Fig. 10a–c illustrate porous grain-packing (each grain with a porosity of 0.2) dissolution under different reactive transport conditions, while Fig. 10d provides a comparison with the packing of impervious grains. Time snapshots are selected for each scenario when the overall volume fraction of calcite reaches 30%. The remaining volume fractions for each grain column are displayed at the top of subfigures.

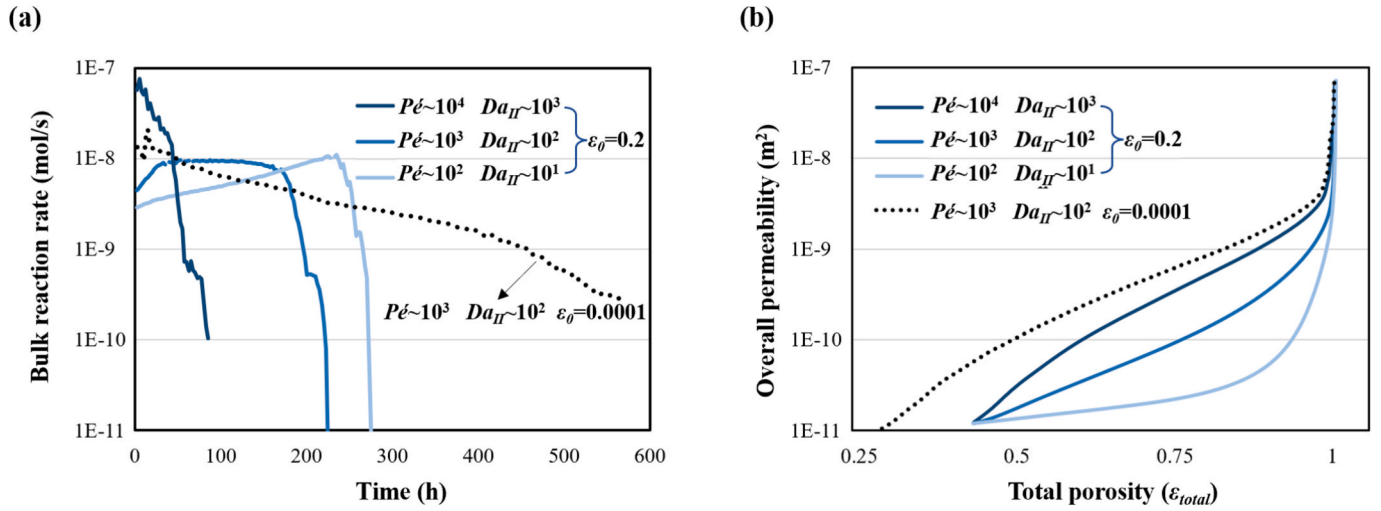


Fig. 11. Comparisons of temporal variations in (a) bulk reaction rate and (d) porosity–permeability relationships across simulation scenarios with different reactive transport regimes and initial intragranular porosity. In Fig. 11b, the x-axis porosity represents the total porosity, and the overall permeability is determined from the total pressure drop across the inlet and outlet boundaries of the entire domain.

of permeability evolution but leads to a progressively diminished extent of grain alteration along the flow direction (shown in Fig. S7 of Text S5). Compared to packings of impermeable grains, the overall permeability can be lower by up to two orders of magnitude at the same porosity for the entire domain.

It is important to note that permeability evolution remains unaffected by reactive transport regimes in impermeable grain packings, even though grain geometry may differ due to varying dissolution patterns (detailed in Fig. S8 of Text S6). On the contrary, permeability enhancement following the reaction varies significantly under different

reactive transport conditions in permeable grain packs. Specifically, permeability evolution is considerably enhanced under advection-dominated conditions at the early stage of the dissolution. Preferential surface reactions contribute to noticeable shrinkage of grains, resulting in permeability increases that are more than 20 times greater than those in grain packings in diffusion-controlled regimes. Porosity and permeability changes are usually described by a power law relationship, $\frac{K}{K_0} = \left(\frac{\varepsilon}{\varepsilon_0}\right)^{\omega}$, where the exponent ω is typically assumed to be 3 in fractured media (Al-Khulaifi et al., 2019). During the early dissolution stage across scenarios, the relationship between normalized porosity and permeability can be fitted by a power law with $f = 4.5$ for impermeable grain packings. For porous cases with decreasing *Pé* and Da_{II} numbers, the corresponding f decreases from 6.4 to 3.5 to 1, as intergranular space enlargement is limited by the delayed shrinkage of individual grains (Hommel et al., 2018; Min et al., 2016).

Experimental investigations have also shown the impacts of microporosity on reaction driven porosity and permeability evolution. Core-flooding experiments were performed using two limestones: Lavoux and Euville (Briolet et al., 2025). The Lavoux limestone is primarily composed of highly micritized ooids of approximately 300 μm, with a total porosity exceeding 25 %. Intragranular (micropore) porosity accounts for more than 50 % of the total pore volume and the micropores and macropores are well-connected. In contrast, the Euville limestone has a lower microporosity and poorer pore interconnectivity. Under reactive transport conditions characterized by low *Pé* ($< 10^2$) and Da_{II} ($< 10^{-2}$) numbers, acid solution was able to infiltrate both intergranular macropores and intragranular micropores in Lavoux limestone, progressively altering the surfaces and internal structures of the ooids during dissolution. Consequently, the permeability increased by approximately one order of magnitude following a ~15 % increase in total porosity in the Lavoux limestone, while the Euville limestone exhibited a substantially greater increase in permeability—up to 40-fold—for a comparable porosity gain of ~15 %. This is consistent with our observations in Fig. 10, in which the coupling between mineral reaction and macropore enlargement is stronger without intragranular porosity. In presence of heterogeneity in macropores as in the referenced experiments, this coupling can better facilitate the formation of the self-organization phenomenon. The experimental observations reiterate our finding that intragranular porosity plays a critical role in controlling the evolution of porous media and the resulting porosity–permeability relationship, and thus neglecting the effects of grain porosity and dissolution regime can lead to significant errors in permeability prediction. Systematic investigations are yet to be performed to formulate generalizable porosity–permeability relationships that can account for intragranular porosity in various grain packings, including those corresponding to tight rocks.

5. Implications and limitations

Intragranular porosity, typically caused by pore structures on the nano- to micrometer scale, are ubiquitous in natural rocks. These fine-scale features can significantly contribute to surface area, pore connectivity, and overall reaction rates. Our simulation results have highlighted the limited extent of increase in the bulk reaction rate and complex temporal changes in surface averaged reaction rate, following the variations in the relative contributions of surface and internal reactions. This confirms previous observations that intragranular pores can serve as diffusion-limited microenvironments and key drivers of biogeochemistry processes (Kessler et al., 2014). Surface area associated with intragranular porosity can be accessed via diffusion, and is ignored by measurements of the geometric surface area (GSA) or X-ray CT imaging. Although BET surface areas include the contribution of intragranular porosity, they do not account for the diffusion limitation and thus may overestimate the reactive surface area. In our simulation results, the bulk reaction rates accounted for both intragranular surface areas and local diffusion constraints. Area-averaged reaction rates normalized by the GSA, which decreases monotonically as the grain shrinks, display the same temporal profiles as the bulk reaction rates, in contrast to those normalized by the total surface area (BET equivalent) (Fig. 3b & Fig. 5b). Notably, the GSA-normalized rates are larger than the BET-normalized values by a factor of five on average, and by more than one order of magnitude at the peak of bulk reaction rates. Therefore, failure to account for the impact of intragranular porosity on reactive surface areas is likely to be one of the contributing factors to the substantial discrepancies between laboratory-derived and field-scale reaction rates, particularly over long-time scales.

Furthermore, our simulations showed that intragranular porosity can result in different textures within the grain under different reactive transport regimes. This means that the textural information recorded in grains may be used to reconstruct reactive transport conditions in geological times. Additionally, grain boundary-connected microporosity is critical for determining where reactions initiate and how they propagate (Briolet et al., 2025), thereby influencing porosity, permeability, and mineral transformations. The insights gained from this study underscore the importance of resolving intragranular features for realistic prediction of fluid–rock interactions in both natural systems (e.g., metasomatism, weathering, and ore genesis) and engineered applications such as CO_2 sequestration, enhanced geothermal systems, and contaminant remediation.

Incorporating such pore-scale mechanisms is needed to understand large-scale and long-term geochemical evolution, but how to upscale such micro-scale dynamics will require further investigations. The hybrid model proposed in our study provides a tool to predict grain scale bulk reaction rate under certain reactive transport conditions, and similar to the shrinking core model, it may be integrated into continuum scale models (MacQuarrie and Mayer, 2005). However, it should be noted that as the shape parameters used in the hybrid model would vary based on local reactive transport environment, parameterization will be more demanding in this case. Similarly, porosity–permeability relationship in grain packs with intragranular porosity takes different forms under varying reactive transport regimes, modifying porosity–permeability relationships to account for the impacts of intragranular porosity may not be straightforward. The shared dependence of bulk reaction and permeability evolution on reactive transport conditions highlights the need to coordinate the upscaling of the coupled flow-transport-reaction processes (Deng et al., 2022). The dual/multi-continua models that have been used for fractured porous media (Doughty, 1999; Lichtner, 2000) may provide a framework for simultaneously accounting for the contrasting properties of intragranular porosity and macro-porosity, provided that the mass exchange term and respective constitutive relations can be properly formulated.

In this study, we focused on quantifying the impacts of different levels of intragranular porosity on the evolution of dissolving porous grains in comparison with impervious grains, and thus considered a single mineral and assumed that intragranular porosity is homogeneous within the grain. The few cases accounting for spatial heterogeneities in microstructures and mineralogical textures are based on simplified conceptualizations and may not fully capture what was expected in real systems, and more systematic investigations are needed in the future. Specifically, the tortuosity and connectivity of intragranular pores can be more complex and variable in reality. The evolution of porous grains in presence of minerals with varying reactivities may be described by the SCM, as the ash layer considered in the model provides an approximation of the remaining layer composed of minerals of low reactivities. Previous studies of fractures bordering multi-mineral porous matrix may also be used to draw some insights, which showed an increasingly limited reaction rate due to diffusion limitation through the remaining layer (Deng et al., 2016; Zhang et al., 2024a). Spatial heterogeneities in microstructures can be resolved in numerical simulations to capture local gradients, but developing generalizable reaction rate models as in

this study may be challenging. One particular type of heterogeneity involves the nano-scale features, future research is necessary to properly account for the nano-confinement effect on fluid and mineral properties (Diao and Espinosa-Marzal, 2016) and the nano-scale variations in reactivity (e.g., at step edge vs. kink sites) (Arvidson et al., 2003).

Although our simulations are 2D, the workflow can be readily extended to 3D. For homogeneous grains, the mathematical equations of the hybrid model can be revised for granular instead of cylindrical shapes with updated geometric formulations. The changes in bulk properties in packings of porous grains observed in 2D are expected to hold in 3D, but future investigations are needed for more realistic packings. Capturing flow and transport in 3D pores between grains accurately is necessary because they dictate local concentration gradients and the accessibility and thus the extent of contributions of the intragranular porosity (Scheibe et al., 2013). The dependence of porous grain evolution on reactive transport regimes defined by varied flow velocity in the open pore space as shown in section 3.2 also highlights the necessity of representing flow truthfully in realistic packings. Additionally, whether the hybrid model can be extended to account for evolution of grains with more complex internal porosity needs to be assessed.

One process that was out of the scope of our study but is extremely important is the mechanical process. As mineral dissolution proceeds, layers of increased internal porosity develop within the grain. It is possible that the highly porous altered layer will become mechanically unstable and disintegrate. This process may expose new reactive surface area, and further enhance reaction (Lee et al., 1998). Previous studies of fracture evolution with an altered layer on the fracture surface have shown the impacts of disaggregation of the porous altered layer on channelization in the fracture and on reaction rate via mitigating the diffusion limitation through the altered layer (Deng et al., 2017). However, the coupling of chemical and mechanical processes in shaping the intragranular porosity evolution at the grain scale warrants further examinations.

6. Conclusions

Intragranular porosity and thus permeable fluid-mineral interface, prevalent in fractured porous media or rocks, can lead to dissolution dynamics that are notably different from impervious cases, and thus should be considered in predicting their bulk properties (e.g., hydraulic properties) and subsequent evolution. In this work, systematic numerical experiments using a micro-continuum modeling framework were conducted at the single-grain scale and the grain packing scale to evaluate the impacts of intragranular porosity, i.e., permeable fluid-mineral interface. By fully resolving the coupled transport and reaction processes at both grain-fluid interfaces and within intragranular domains, our study quantitatively assessed the competition between surface and internal reactions, demonstrating that this balance is strongly governed by the initial intragranular porosity and the prevailing reactive transport regime.

In contrast to the interface-controlled alterations of impervious grains, mass transfer through porous grain boundaries enabled internal reactions, especially with well-connected intragranular porosity, which can account for a majority of the total reaction and contributes to faster overall dissolution. The bulk dissolution rate increases with intragranular porosity first and decreases at larger intragranular porosities, as a result of the competing effect of reactive area and depletion of reactive fluids. Over time, bulk dissolution rate increases and then decreases following the temporal trends of reactive areas resulting from the development of more porous zones within the grains. Compared to the impervious cases, surface reaction is dampened and thus changes in both the size and shape of the grain are delayed. These effects are particularly pronounced in highly porous grains, e.g., initial intragranular porosity exceeding 0.2, and under diffusion-controlled conditions, i.e., at low $Pé$ and Da_{II} numbers. Furthermore, the evolution of grain internal texture becomes increasingly complex when microcrystal-scale heterogeneities are considered, which may arise from variations in microcrystal size, packing arrangement, and spatial distribution.

Distinct dissolution features, as characterized by grain shape, equivalent diameter and average density, were identified at different stages of the dissolution, as the relative contribution of surface reaction and internal reaction changes. These features can be used to parameterize a hybrid conversion model we proposed, which assumes a uniformly interior reaction regime and can successfully reproduce the bulk dissolution rate dynamics. Moreover, the proposed hybrid model can be incorporated into continuum-scale simulations, but requesting additional parameterization of shape indicators to account for varying reactive transport conditions.

The impacts of porous grains on reactive transport behaviors can be amplified at the grain packing scale, i.e., in bulk porous media. The restricted shrinkage of mineral grains and larger dissolution rate caused by enhanced internal reactions would result in faster bulk porosity increase. However, for the same porosity increase, the permeability increase can be up to two orders of magnitude lower than the packing of impervious grains. The trajectory of permeability enhancement is insensitive to reactive transport conditions in impermeable grain packings, whereas permeability evolution of porous grain packings deviate from the impervious case more significantly at lower $Pé$ and Da_{II} numbers. This behavior arises from the strong dependence of the relative contribution of internal reactions on the reactive transport regimes.

Our findings underscore the necessity of accounting for sub-resolution structural and mineralogical features for accurate long-term predictions, and offer important insights for engineering applications to formulate more suitable bulk reaction progression and porosity–permeability relationships to properly account for features such as intragranular porosity. Although this study focuses on a simple reactive transport system, the micro-continuum approach can be extended to other systems with increasing extent of rock heterogeneity in pore structure and mineralogy. Our future research will aim to capture changes in morphological and hydrodynamic properties of heterogeneous porous media, and the feedback with geochemical reactions with coupling of mechanical processes for broader applications.

CRediT authorship contribution statement

Qian Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hang Deng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Yanhui Dong:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Data availability

Simulation results and setup files can be found online at Zenodo via https://doi.org/10.5281/zenodo.16918722. The opensource code used can be downloaded at: https://github.com/cisteefel/crunchtope.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary Material to this paper provides additional results about the validation of the micro-continuum reactive transport model (Text S1, Fig. S1-S2), simulation scenario conceptualization and setup (Text S2, Fig. S3-S4 and Table S1), spatial distribution of area averaged reaction rates for single grain cases (Text S3 and Fig. S5), and statistical analyses of parameters of the proposed hybrid conversion model (Text S4 and Fig. S6). Detailed analyses of permeability evolution are also available for cases with lower flow velocities (Text S5 and Fig. S7) or impermeable fluid–solid interfaces under various reactive transport regimes (Text S6 and Fig. S8). Supplementary material to this article can be found online at https://doi.org/10.1016/j.gca.2025.11.030.

References

Ahn, H., Choi, S., 2017. A comparison of the shrinking core model and the grain model for the iron ore pellet indurator simulation. Comput. Chem. Eng. 97, 13–26.

Al-Khulaifi, Y., Lin, Q., Blunt, M.J., Bijeljic, B., 2017. Reaction rates in chemically heterogeneous rock: coupled impact of structure and flow properties studied by X-ray microtomography. Environ. Sci. Technol. 51 (7), 4108–4116.

Al-Khulaifi, Y., Lin, Q., Blunt, M.J., Bijeljic, B., 2018. Reservoir-condition pore-scale imaging of dolomite reaction with supercritical CO_2 acidified brine: effect of pore-structure on reaction rate using velocity distribution analysis. Int. J. Greenhouse Gas Control 68, 99–111.

Al-Khulaifi, Y., Lin, Q., Blunt, M.J., Bijeljic, B., 2019. Pore-scale dissolution by CO2 saturated brine in a multimineral carbonate at reservoir conditions: Impact of physical and chemical heterogeneity. Water Resour. Res. 55 (4), 3171–3193.

Arvidson, R.S., Ertan, I.E., Amonette, J.E., Luttge, A., 2003. Variation in calcite dissolution rates:: a fundamental problem. Geochim. Cosmochim. Acta 67 (9), 1623–1634.

Batchelor, M.T., Burne, R.V., Henry, B.I., Li, F., Paul, J., 2018. A biofilm and organomineralisation model for the growth and limiting size of ooids. Sci. Rep. 8 (1), 559.

Beckingham, L.E., 2017. Evaluation of macroscopic porosity-permeability relationships in heterogeneous mineral dissolution and precipitation scenarios. Water Resour. Res. 53 (12), 10217–10230.

Brantley, S.L., 2008. Kinetics of mineral dissolution. In: Kinetics of Water-Rock Interaction. New York, NY, Springer, New York, pp. 151–210.

Briolet, T., Bemer, E., Sissmann, O., Fortin, J., 2025. Localized versus distributed dissolution in carbonate rocks: the key role of microstructure. Geochim. Cosmochim. Acta 397, 113–133.

Carroll, S., Carey, J.W., Dzombak, D., Huerta, N.J., Li, L., Richard, T., Zhang, L., 2016. Role of chemistry, mechanics, and transport on well integrity in CO2 storage environments. Int. J. Greenhouse Gas Control 49, 149–160.

Chou, L.E.I., Garrels, R.M., Wollast, R., 1989. Comparative study of the kinetics and mechanisms of dissolution of carbonate minerals. Chem. Geol. 78 (3–4), 269–282.

Deng, H., Molins, S., Steefel, C., DePaolo, D., Voltolini, M., Yang, L., & Ajo-Franklin, J. (2016). A 2.5 D reactive transport model for fracture alteration simulation. Environmental science & technology, 50(14), 7564-7571.Deng, H., & Spycher, N. (2019). Modeling reactive transport processes in fractures. Reviews in Mineralogy and Geochemistry, 85(1), 49-74.

Deng, H., Voltolini, M., Molins, S., Steefel, C., DePaolo, D., Ajo-Franklin, J., Yang, L., 2017. Alteration and erosion of rock matrix bordering a carbonate-rich shale fracture. Environ. Sci. Technol. 51 (15), 8861–8868.

Diao, Y., Espinosa-Marzal, R.M., 2016. Molecular insight into the nanoconfined calcite–solution interface. Proc. Natl. Acad. Sci. 113 (43), 12047–12052.

Doraiswamy, L.K., 1984. Heterogeneous reactions: analysis, examples, and reactor design. Gas-Solid Solid-Solid React. 1, 43.

Doughty, C., 1999. Investigation of conceptual and numerical approaches for evaluating moisture, gas, chemical, and heat transport in fractured unsaturated rock. J. Contam. Hydrol. 38 (1–3), 69–106.

Dutka, F., Starchenko, V., Osselin, F., Magni, S., Szymczak, P., Ladd, A.J., 2020. Time-dependent shapes of a dissolving mineral grain: comparisons of simulations with microfluidic experiments. Chem. Geol. 540, 119459.

Ewing, R.P., Hu, Q., Liu, C., 2010. Scale dependence of intragranular porosity, tortuosity, and diffusivity. Water Resour. Res. 46 (6).

Gerardo, S., Matthews, K., Warner, J., Song, W., 2023. Role of nanoscale crystallinity on the recovery of rare earth elements (REEs) from coal fly ash. Environ. Sci. Technol. Lett. 10 (10), 943–948.

Gharbi, O., Blunt, M.J., 2012. The impact of wettability and connectivity on relative permeability in carbonates: a pore network modeling analysis. Water Resour. Res. 48 (12).

Gómez-Barea, A., Leckner, B., 2010. Modeling of biomass gasification in fluidized bed. Prog. Energy Combust. Sci. 36 (4), 444–509.

Gray, F., Anabaraonye, B.U., Crawshaw, J.P., Boek, E.S., 2021. Pore-scale dissolution mechanisms in calcite-CO2-brine systems: the impact of non-linear reaction kinetics and coupled ion transport. Geochim. Cosmochim. Acta 305, 323–338.

Halama, S., Spliethoff, H., 2015. Numerical simulation of entrained flow gasification: Reaction kinetics and char structure evolution. Fuel Process. Technol. 138, 314–324.

Hashim, M.S., Kaczmarek, S.E., 2019. A review of the nature and origin of limestone microporosity. Mar. Pet. Geol. 107, 527–554.

Hay, M.B., Stoliker, D.L., Davis, J.A., Zachara, J.M., 2011. Characterization of the intragranular water regime within subsurface sediments: Pore volume, surface area, and mass transfer limitations. Water Resour. Res. 47 (10).

Hommel, J., Coltman, E., Class, H., 2018. Porosity–permeability relations for evolving pore space: a review with a focus on (bio-) geochemically altered porous media. Transp. Porous Media 124 (2), 589–629.

Janjuhah, H.T., Alansari, A., Ghosh, D.P., Bashir, Y., 2018. New approach towards the classification of microporosity in Miocene carbonate rocks, Central Luconia, offshore Sarawak, Malaysia. J. Natl. Gas Geosci. 3 (3), 119–133.

Julia, M., Putnis, C.V., King, H.E., Renard, F., 2023. Coupled dissolution-precipitation and growth processes on calcite, aragonite, and Carrara marble exposed to cadmium-rich aqueous solutions. Chem. Geol. 621, 121364.

Kaczmarek, S.E., Fullmer, S.M., Hasiuk, F.J., 2015. A universal classification scheme for the microcrystals that host limestone microporosity. J. Sediment. Res. 85 (10), 1197–1212.

Kelemen, P.B., McQueen, N., Wilcox, J., Renforth, P., Dipple, G., Vankeuren, A.P., 2020. Engineered carbon mineralization in ultramafic rocks for CO2 removal from air: Review and new insights. Chem. Geol. 550, 119628.

Kessler, A.J., Cardenas, M.B., Santos, I.R., Cook, P.L., 2014. Enhancement of denitrification in permeable carbonate sediment due to intra-granular porosity: a multi-scale modelling analysis. Geochim. Cosmochim. Acta 141, 440–453.

Kopanja, L., Žunić, D., Lončar, B., Gyergyek, S., Tadić, M., 2016. Quantifying shapes of nanoparticles using modified circularity and ellipticity measures. Measurement 92, 252–263.

Lee, M.R., Hodson, M.E., Parsons, I., 1998. The role of intragranular microtextures and microstructures in chemical and mechanical weathering: direct comparisons of experimentally and naturally weathered alkali feldspars. Geochim. Cosmochim. Acta 62 (16), 2771–2788.

Li, L., Steefel, C.I., Yang, L., 2008. Scale dependence of mineral dissolution rates within single pores and fractures. Geochim. Cosmochim. Acta 72 (2), 360–377.

Lichtner, P. C. (2000). Critique of dual continuum formulations of multicomponent reactive transport in fractured porous media (No. LA-UR-00-1097). Los Alamos National Lab.(LANL), Los Alamos, NM (United States).

Liu, M., Shabaninejad, M., Mostaghimi, P., 2017. Impact of mineralogical heterogeneity on reactive transport modelling. Comput. Geosci. 104, 12–19.

Liu, M., Starchenko, V., Anovitz, L.M., Stack, A.G., 2020. Grain detachment and transport clogging during mineral dissolution in carbonate rocks with permeable grain boundaries. Geochim. Cosmochim. Acta 280, 202–220.

Luo, H., Quintard, M., Debenest, G., Laouafa, F., 2012. Properties of a diffuse interface model based on a porous medium theory for solid–liquid dissolution problems. Comput. Geosci. 16, 913–932.

Ma, L. (2006). Combustion and gasification of chars in oxygen and carbon dioxide at elevated pressure (Doctoral dissertation, Stanford University).

MacQuarrie, K.T., Mayer, K.U., 2005. Reactive transport modeling in fractured rock: a state-of-the-science review. Earth Sci. Rev. 72 (3–4), 189–227.

Maher, K., Navarre-Sitchler, A., 2019. Reactive transport processes that drive chemical weathering: from making space for water to dismantling continents. Rev. Mineral. Geochem. 85 (1), 349–380.

Maher, K., Steefel, C.I., DePaolo, D.J., Viani, B.E., 2006. The mineral dissolution rate conundrum: Insights from reactive transport modeling of U isotopes and pore fluid chemistry in marine sediments. Geochim. Cosmochim. Acta 70 (2), 337–363.

Mayo, S., Josh, M., Nesterets, Y., Esteban, L., Pervukhina, M., Clennell, M.B., Hall, C., 2015. Quantitative micro-porosity characterization using synchrotron micro-CT and xenon K-edge subtraction in sandstones, carbonates, shales and coal. Fuel 154, 167–173.

Min, T., Gao, Y., Chen, L., Kang, Q., Tao, W.W., 2016. Changes in porosity, permeability and surface area during rock dissolution: effects of mineralogical heterogeneity. Int. J. Heat Mass Transf. 103, 900–913.

Molins, S., 2015. Reactive interfaces in direct numerical simulation of pore-scale processes. Rev. Mineral. Geochem. 80 (1), 461–481.

Molins, S., Knabner, P., 2019. Multiscale approaches in reactive transport modeling. Rev. Mineral. Geochem. 85 (1), 27–48.

Molins, S., Trebotich, D., Arora, B., Steefel, C.I., Deng, H., 2019. Multi-scale model of reactive transport in fractured media: diffusion limitations on rates. Transp. Porous Media 128 (2), 701–721.

Molins, S., Soulaine, C., Prasianakis, N.I., Abbasi, A., Poncet, P., Ladd, A.J., Steefel, C.I., 2021. Simulation of mineral dissolution at the pore scale with evolving fluid-solid interfaces: review of approaches and benchmark problem set. Comput. Geosci. 25, 1285–1318.

Molins, S., Trebotich, D., Miller, G.H., Steefel, C.I., 2017. Mineralogical and transport controls on the evolution of porous media texture using direct numerical simulation. Water Resour. Res. 53 (5), 3645–3661.

Moshier, S.O., 1989. Microporosity in micritic limestones: a review. Sed. Geol. 63 (3–4), 191–213.

Nguyen, C.B., Farid, M.M., Scherer, J., Guo, Q., Gräbner, M., Richter, A., 2022. A hybrid particle model with advanced conversion parameters and dynamic drag model applied for the CFD modeling of an entrained-flow gasifier. Combust. Flame 240, 112040.

Nguyen, C.B., Scherer, J., Hartwich, M., Richter, A., 2021. The morphology evolution of char particles during conversion processes. Combust. Flame 226, 117–128.

Noiriel, C., Daval, D., 2017. Pore-scale geochemical reactivity associated with CO2 storage: new frontiers at the fluid–solid interface. Acc. Chem. Res. 50 (4), 759–768.

Noiriel, C., Luquot, L., Madé, B., Raimbault, L., Gouze, P., van Der Lee, J., 2009. Changes in reactive surface area during limestone dissolution: an experimental and modelling study. Chem. Geol. 265 (1–2), 160–170.
Noiriel, C., Soulaine, C., 2021. Pore-scale imaging and modelling of reactive flow in evolving porous media: Tracking the dynamics of the fluid–rock interface. Transp. Porous Media 140 (1), 181–213.
Popovič, Ľ., Raschman, P., Spišák, J., Fedoročková, A., Sučik, G., 2020. Generalized shrinking particle model of leaching: effect of the order of surface chemical reaction, liquid-to-solid ratio and non-ideal behaviour of the liquid phase. Hydrometall. 196, 105441.
Rücker, M., Bartels, W.B., Singh, K., Brussee, N., Coorn, A., van der Linde, H.A., Berg, S., 2019. The effect of mixed wettability on pore-scale flow regimes based on a flooding experiment in Ketton limestone. Geophys. Res. Lett. 46 (6), 3225–3234.
Salmi, T., Grénman, H., Wärnå, J., Murzin, D.Y., 2013. New modelling approach to liquid–solid reaction kinetics: from ideal particles to real particles. Chem. Eng. Res. Des. 91 (10), 1876–1889.
Scheibe, T.D., Hou, Z., Palmer, B.J., Tartakovsky, A.M., 2013. Pore-scale simulation of intragranular diffusion: Effects of incomplete mixing on macroscopic manifestations. Water Resour. Res. 49 (7), 4277–4294.
Schultz, L.N., Dideriksen, K., Lakshtanov, L., Hakim, S.S., Muter, D., Haußer, F., Stipp, S. L.S., 2014. From nanometer aggregates to micrometer crystals: Insight into the coarsening mechanism of calcite. Cryst. Growth Des. 14 (2), 552–558.
Soulaine, C., Tchelepi, H.A., 2016. Micro-continuum approach for pore-scale simulation of subsurface processes. Transp. Porous Media 113 (3), 431–456.
Soulaine, C., Maes, J., Roman, S., 2021. Computational microfluidics for geosciences. Front. Water 3, 643714.
Soulaine, C., Roman, S., Kovscek, A., Tchelepi, H.A., 2017. Mineral dissolution and wormholing from a pore-scale perspective. J. Fluid Mech. 827, 457–483.
Steefel, C.I., Beckingham, L.E., Landrot, G., 2015. Micro-continuum approaches for modeling pore-scale geochemical processes. Rev. Mineral. Geochem. 80 (1), 217–246.
Wen, H., Li, L., 2017. An upscaled rate law for magnesite dissolution in heterogeneous porous media. Geochim. Cosmochim. Acta 210, 289–305.
Wood, W.W., Kraemer, T.F., Hearn Jr, P.P., 1990. Intragranular diffusion: an important mechanism influencing solute transport in clastic aquifers? Science 247 (4950), 1569–1572.
Zachara, J., Brantley, S., Chorover, J., Ewing, R., Kerisit, S., Liu, C., Stack, A.G., 2016. Internal domains of natural porous media revealed: critical locations for transport, storage, and chemical reaction. Environ. Sci. Technol. 50 (6), 2811–2829.
Zhang, F., Wang, X., Wang, B., Lou, X., Lipiński, W., 2022a. Experimental and numerical analysis of CO2 and CH4 hydrate formation kinetics in microparticles: a comparative study based on shrinking core model. Chem. Eng. J. 446, 137247.
Zhang, Q., Deng, H., Dong, Y., Molins, S., Li, X., Steefel, C., 2022b. Investigation of coupled Processes in Fractures and the Bordering Matrix via a Micro-Continuum Reactive Transport Model. Water Resour. Res. 58 (2), e2021WR030578.
Zhang, Q., Dong, Y., Molins, S., Deng, H., 2024a. The impacts of micro-porosity and mineralogical texture on fractured rock alteration. Water Resour. Res. 60 (6), e2023WR036266.
Zhang, Z., Zheng, K., Chen, L., Yuan, Q., 2024b. Review on accelerated carbonation of calcium-bearing minerals: Carbonation behaviors, reaction kinetics, and carbonation efficiency improvement. J. Build. Eng. 108826.